

Alberts • Johnson • Lewis • Morgan • Raff • Roberts • Walter

Molecular Biology of the Cell

Seventh Edition

Chapter 2

Cell Chemistry and Bioenergetics

Chapter 14

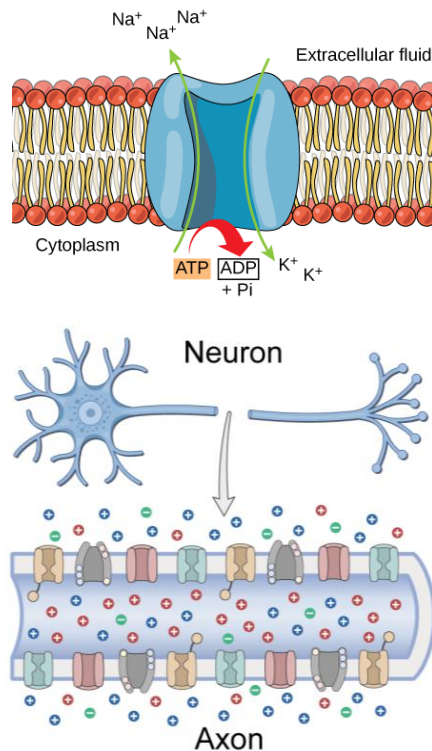
Energy Conversion:

Mitochondria and ~~Chloroplasts~~

Why energy is critical in cell survival?

Numerous biological processes are dependent on energy (ATP)

Ion channels



Anabolism

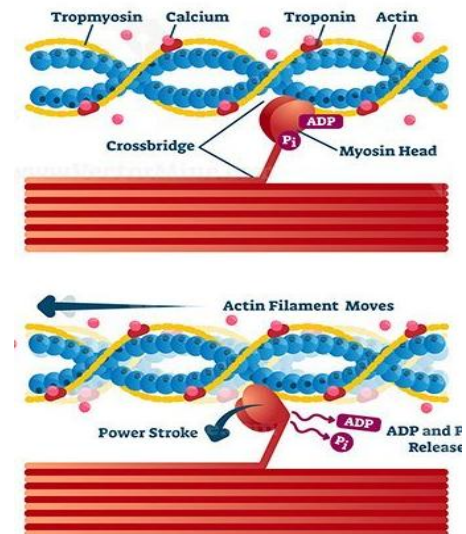
Protein synthesis: enzymes, skeletal fibers, etc.

Lipid synthesis: membranes, fat stores, etc.

Hormones: insulin, estrogen, etc.

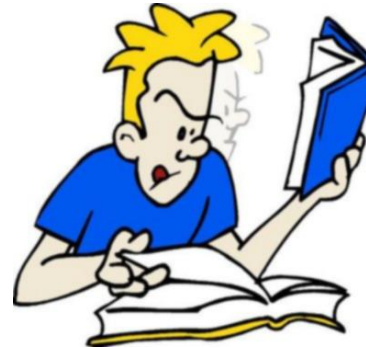
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Muscle contraction



Heart, Skeletal muscles, Smooth muscles, etc.

Biological energy ATP



Average ATP consumption per day :

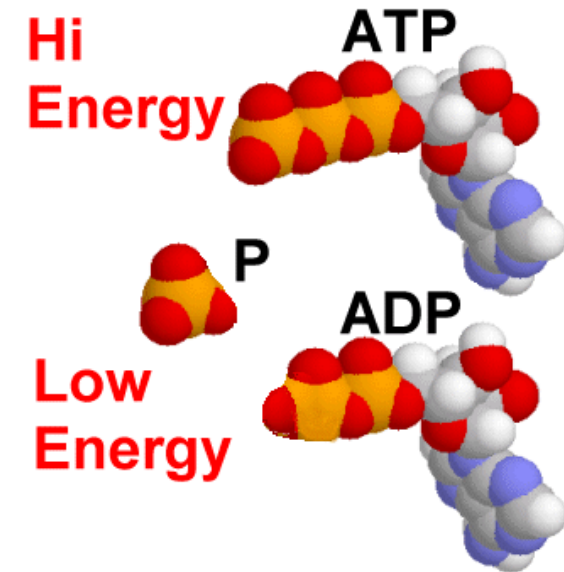
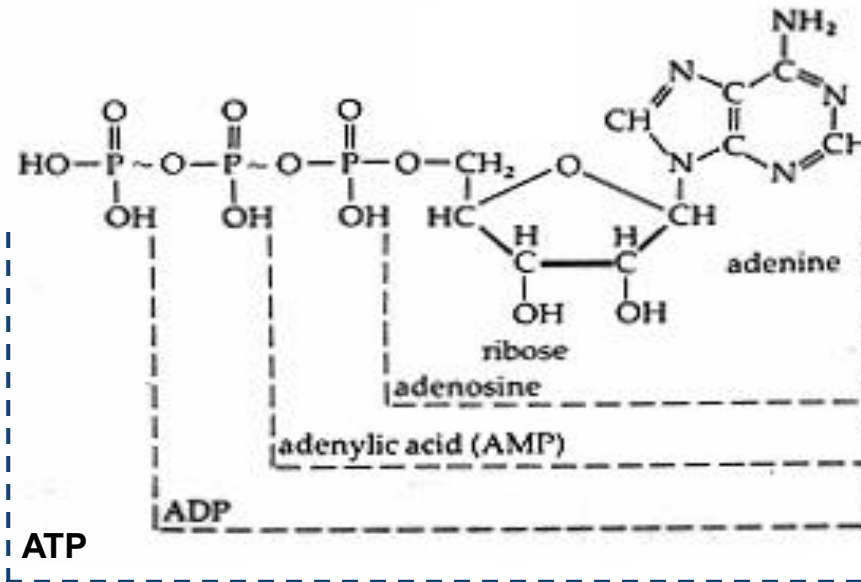
65 kg ATP/70 kg Body weight



What is the energy in biological systems and how does it work?

How the energy is generated?

Biological energy ATP



ATP : Adenosine-Tri-Phospate

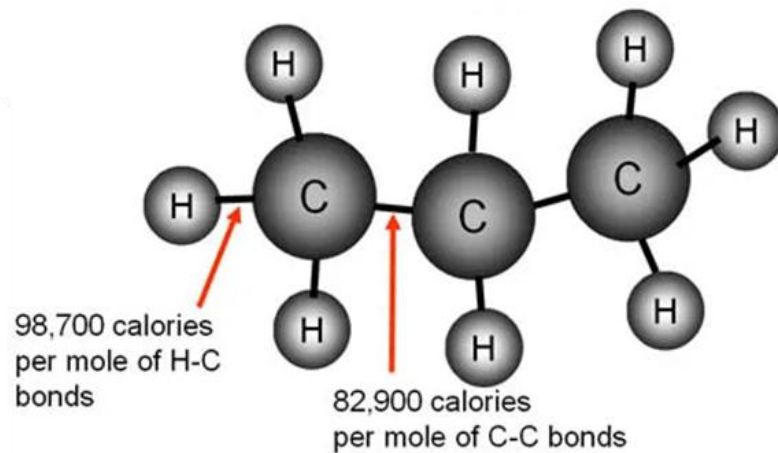
ADP : Adenosine-Di-Phospate

AMP : Adenosine-Mono-Phospate

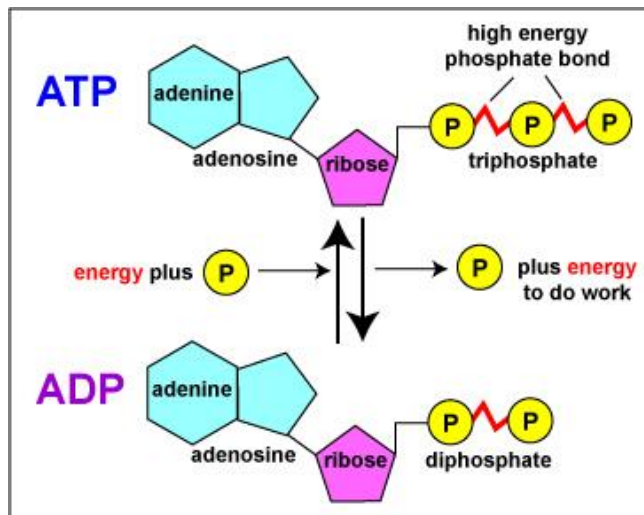
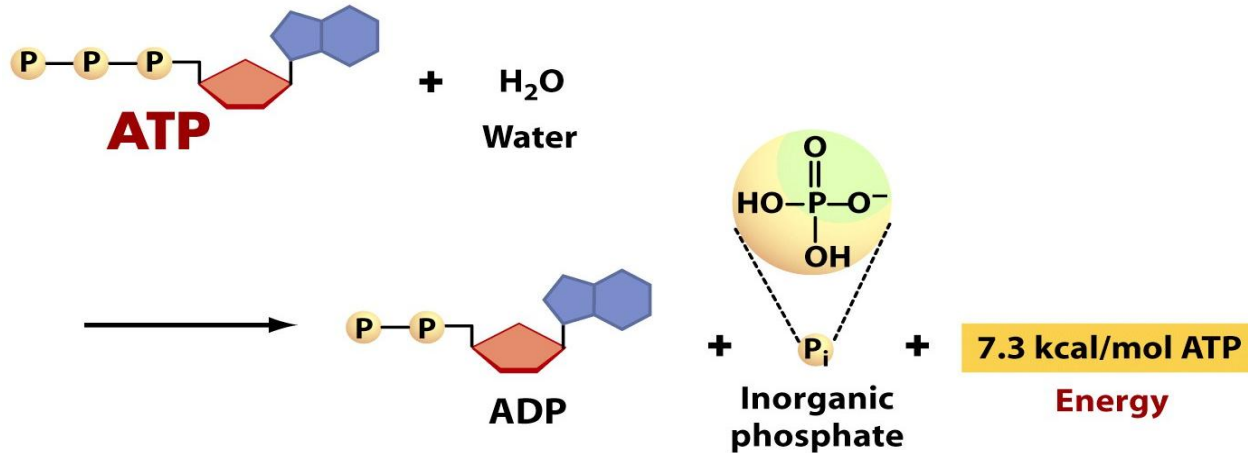
Complex molecules have higher energy level

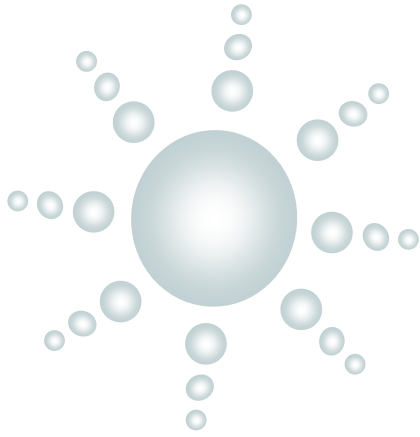
Bond energy

Energy required to maintain chemical bond



Biological energy ATP





***How ATP is used as
biological energy?***

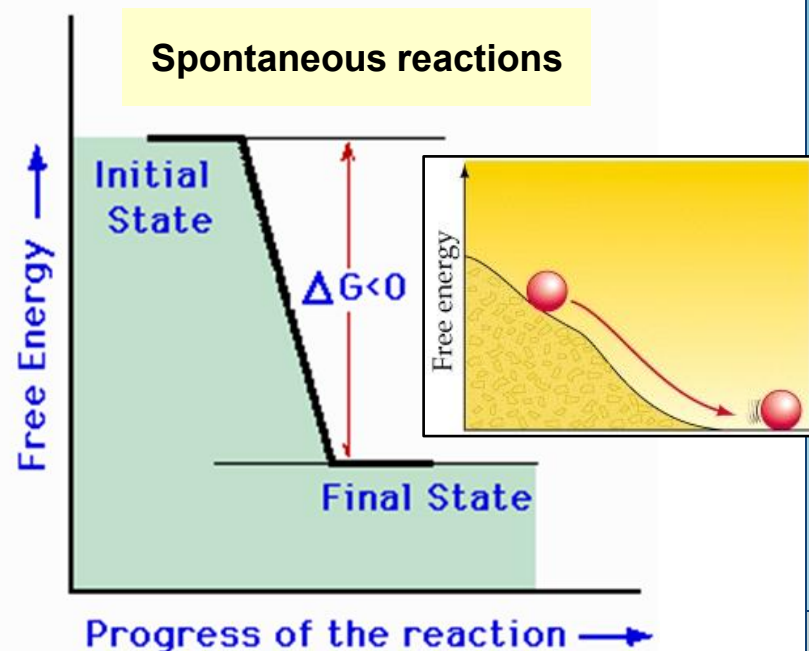
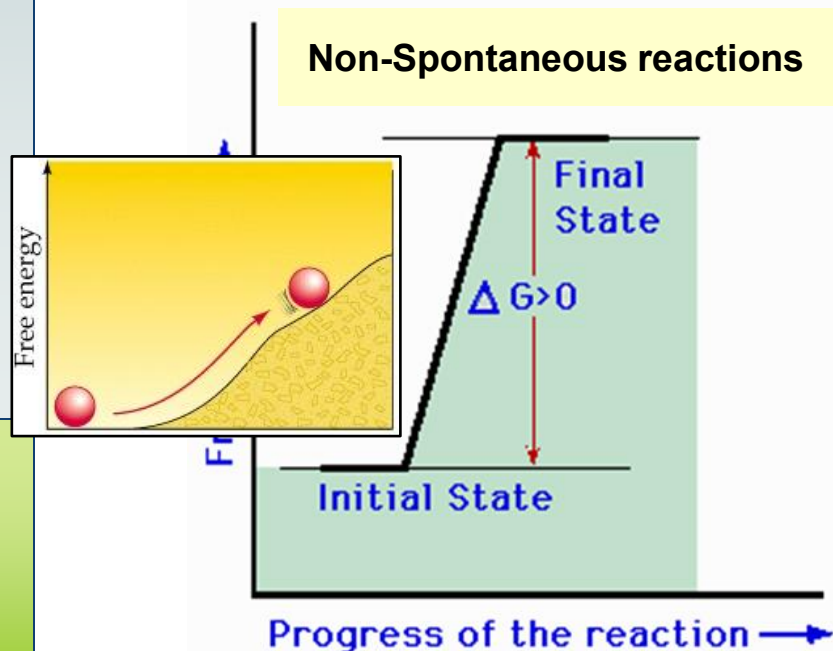
Spontaneous and non-spontaneous reactions

Spontaneous reaction

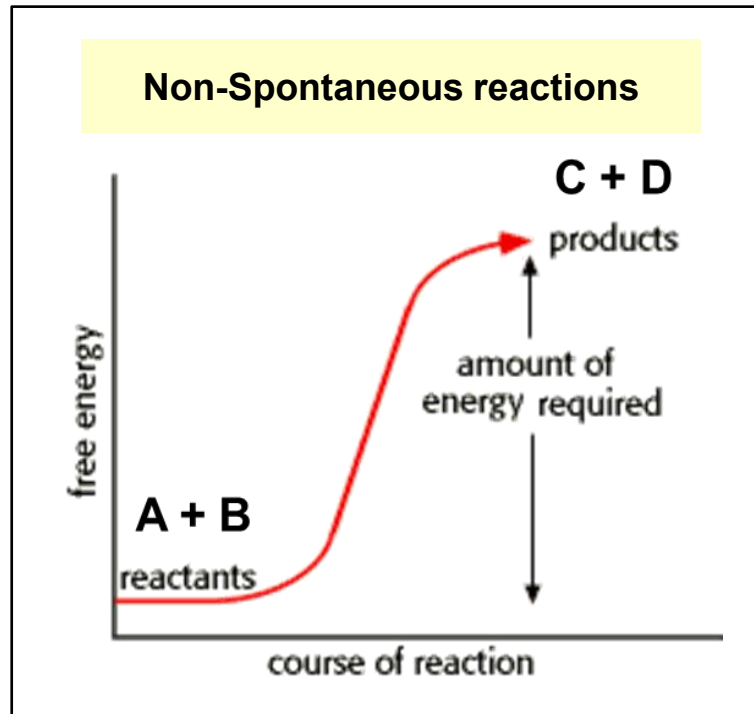
Chemical reaction which decrease energy level of products

Non-spontaneous reaction

Chemical reaction which increase energy level of products



Most of biological reactions are non-spontaneous



Biological reactions

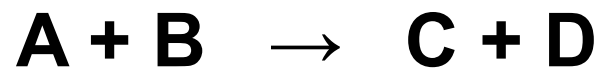
Protein synthesis

DNA synthesis

Hormone synthesis

Physical activities

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·
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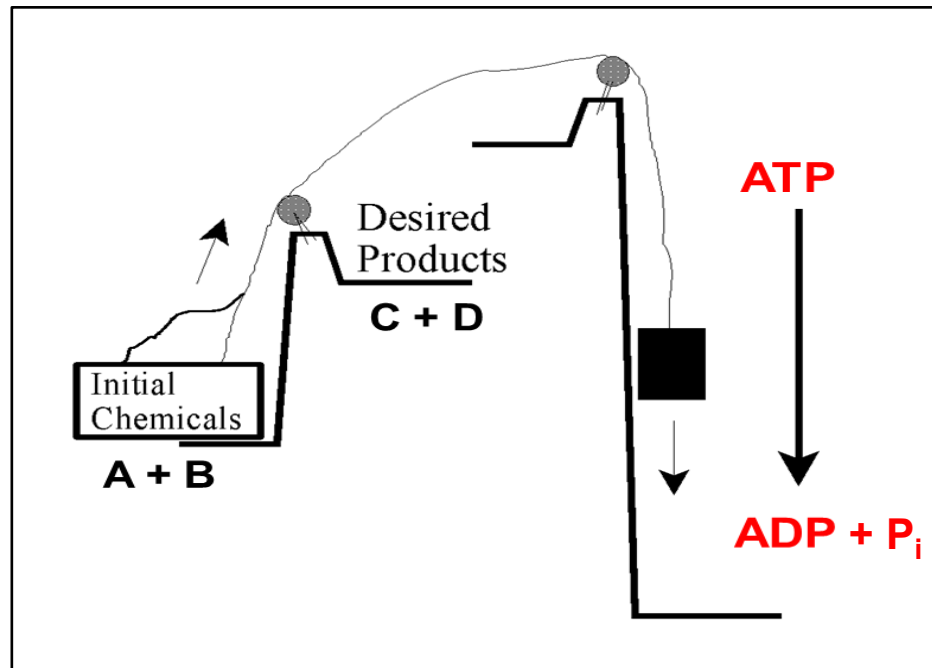


+ Energy

Spontaneous biological reactions

ATP hydrolysis makes biological reaction spontaneous

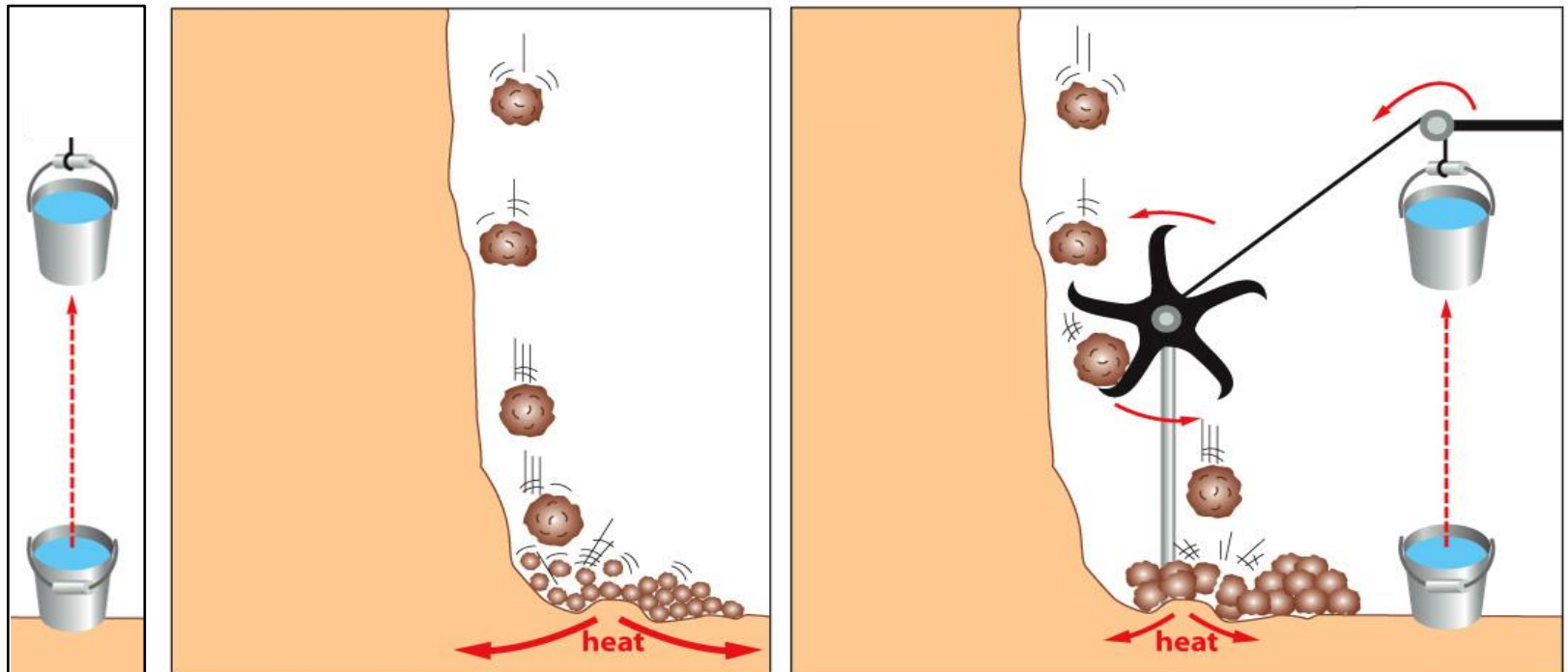
Combination of two reactions through mediator



Spontaneous biological reactions

ATP hydrolysis makes biological reaction spontaneous

Combination of two reactions through mediator

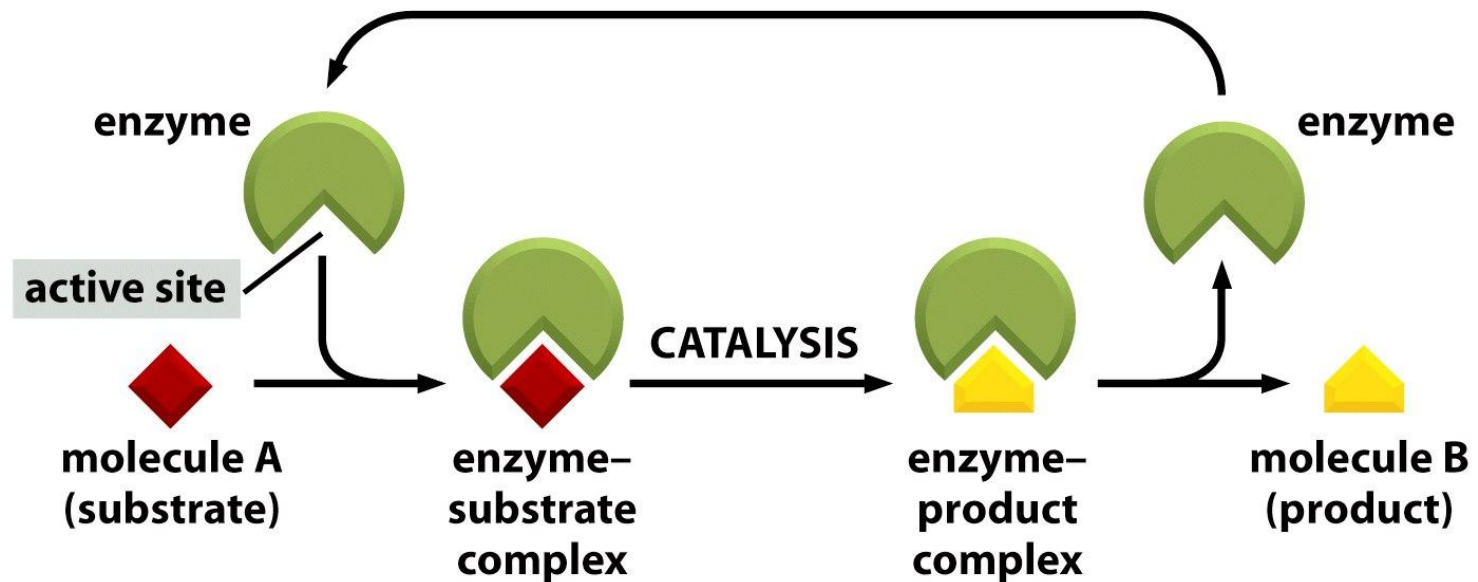


Biological energy ATP



Enzyme (or protein) combine ATP hydrolysis and non-spontaneous reaction

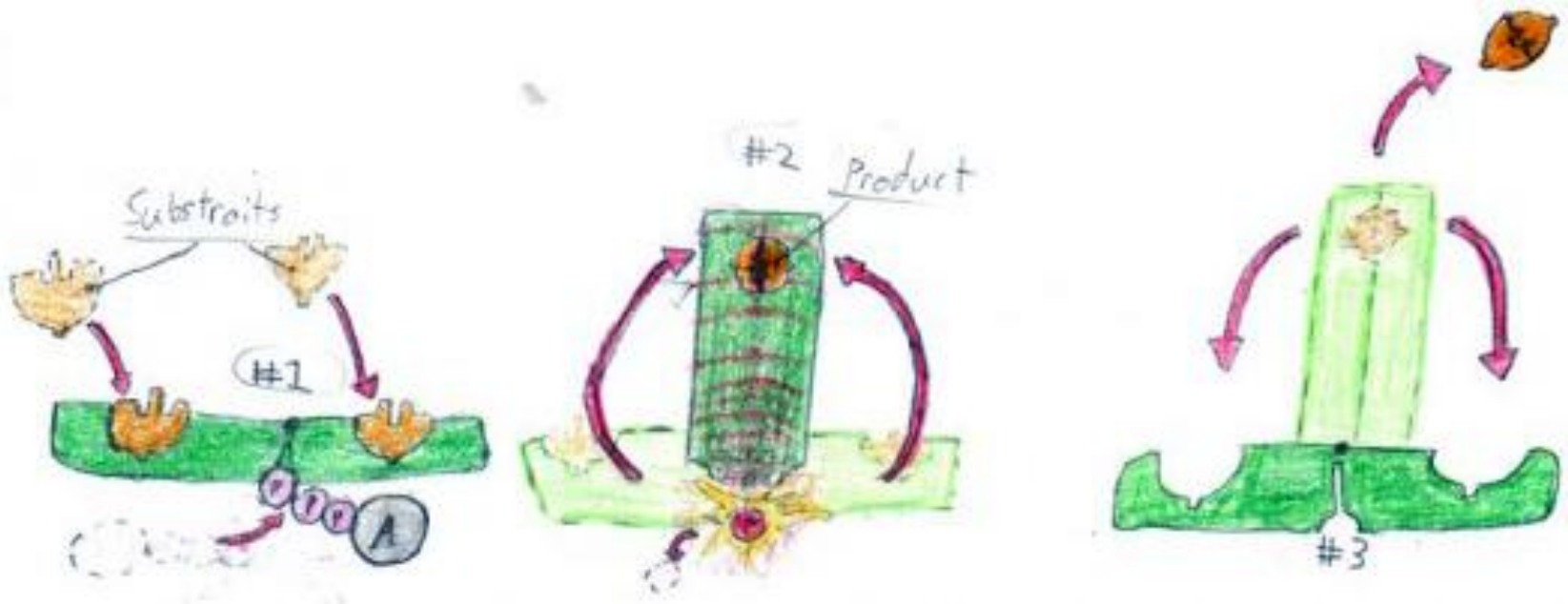
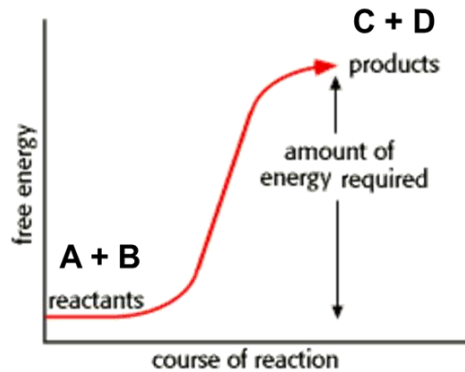
Combination of two reactions by enzyme (or protein)



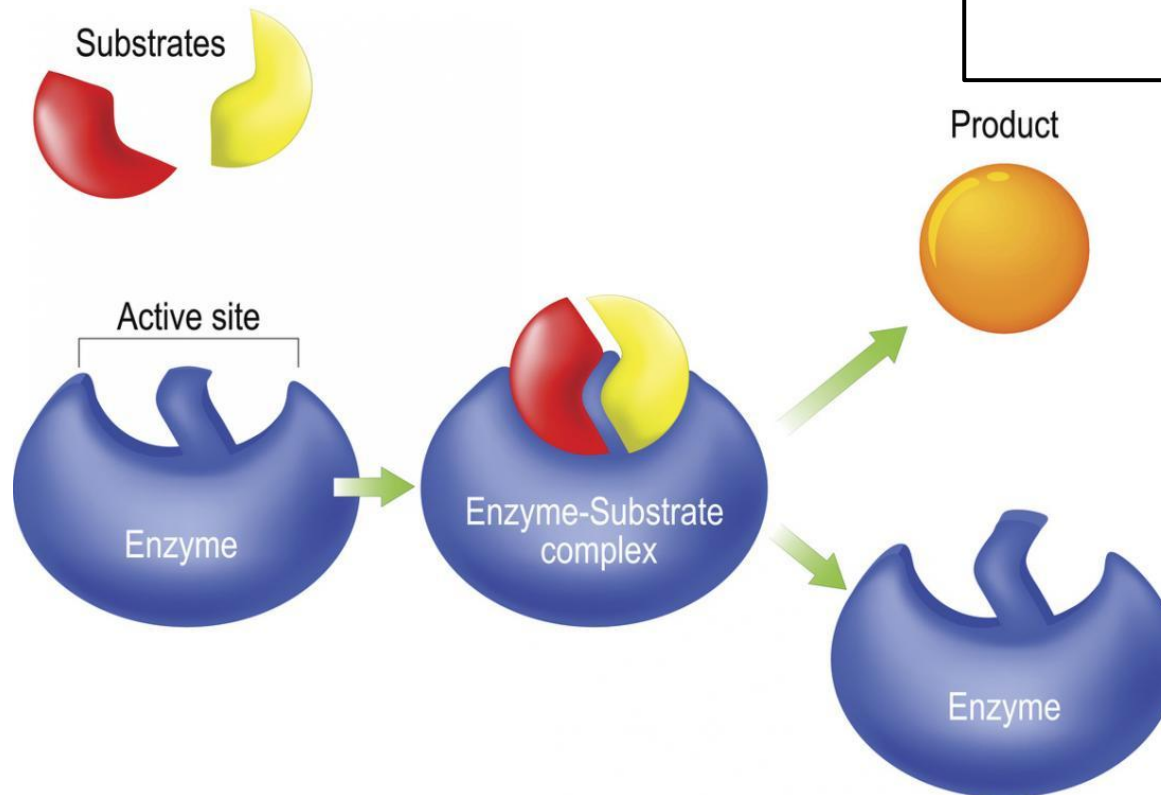
Enzyme reactions (synthesis)



Non-Spontaneous reactions



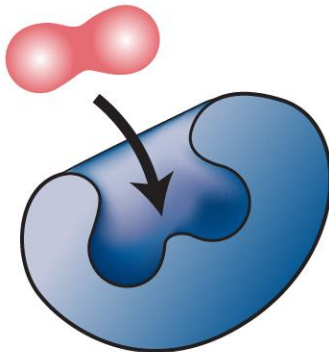
Enzyme reactions (synthesis)



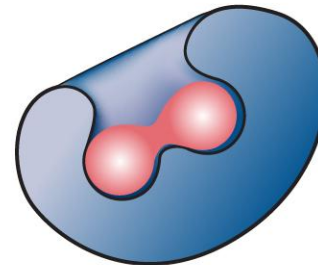
Enzyme reactions (degradation)



Substrate

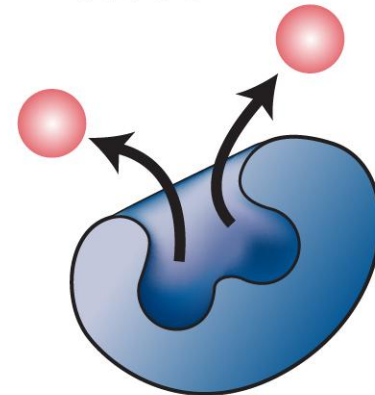


Enzyme



Enzyme-substrate
complex

Products

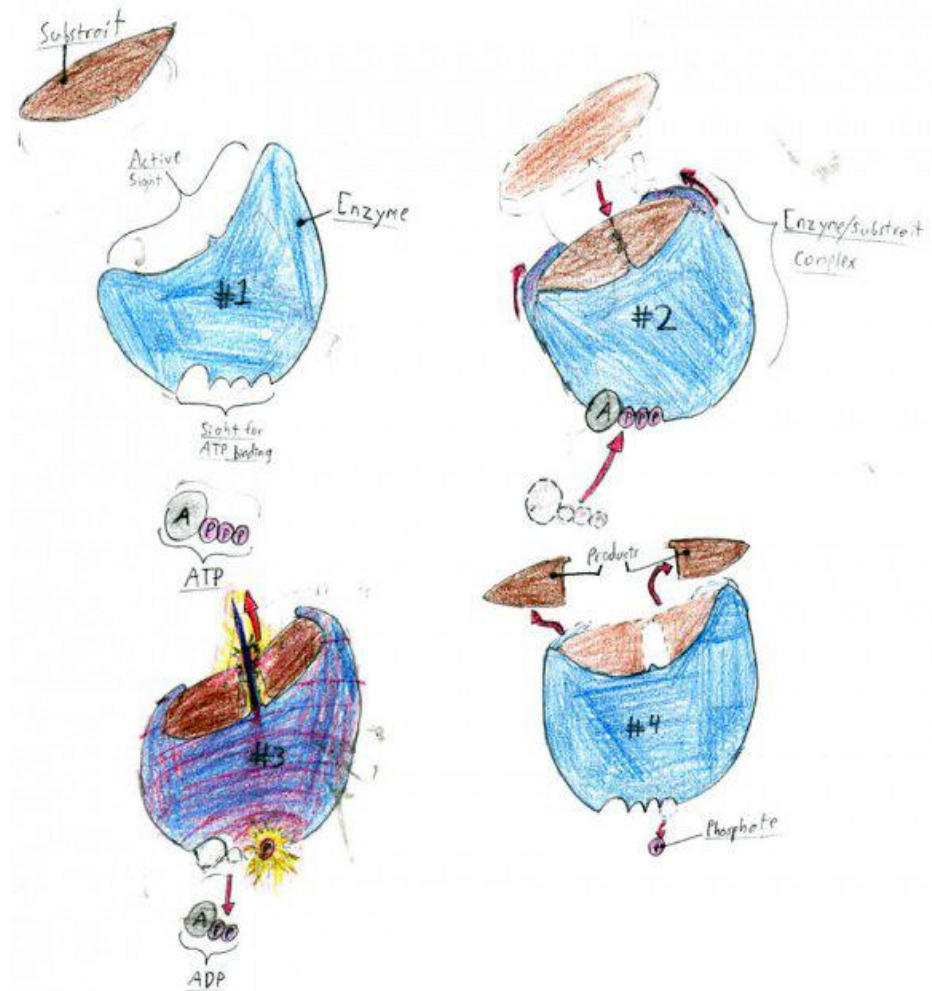
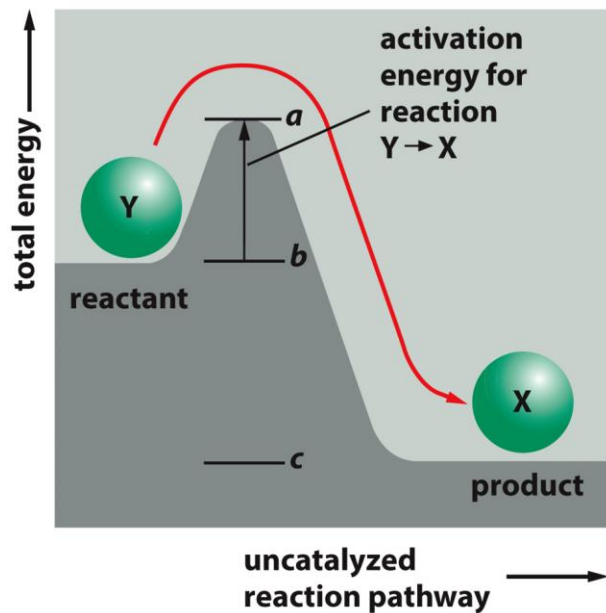


Enzyme

Enzyme reactions (degradation)

Activation energy

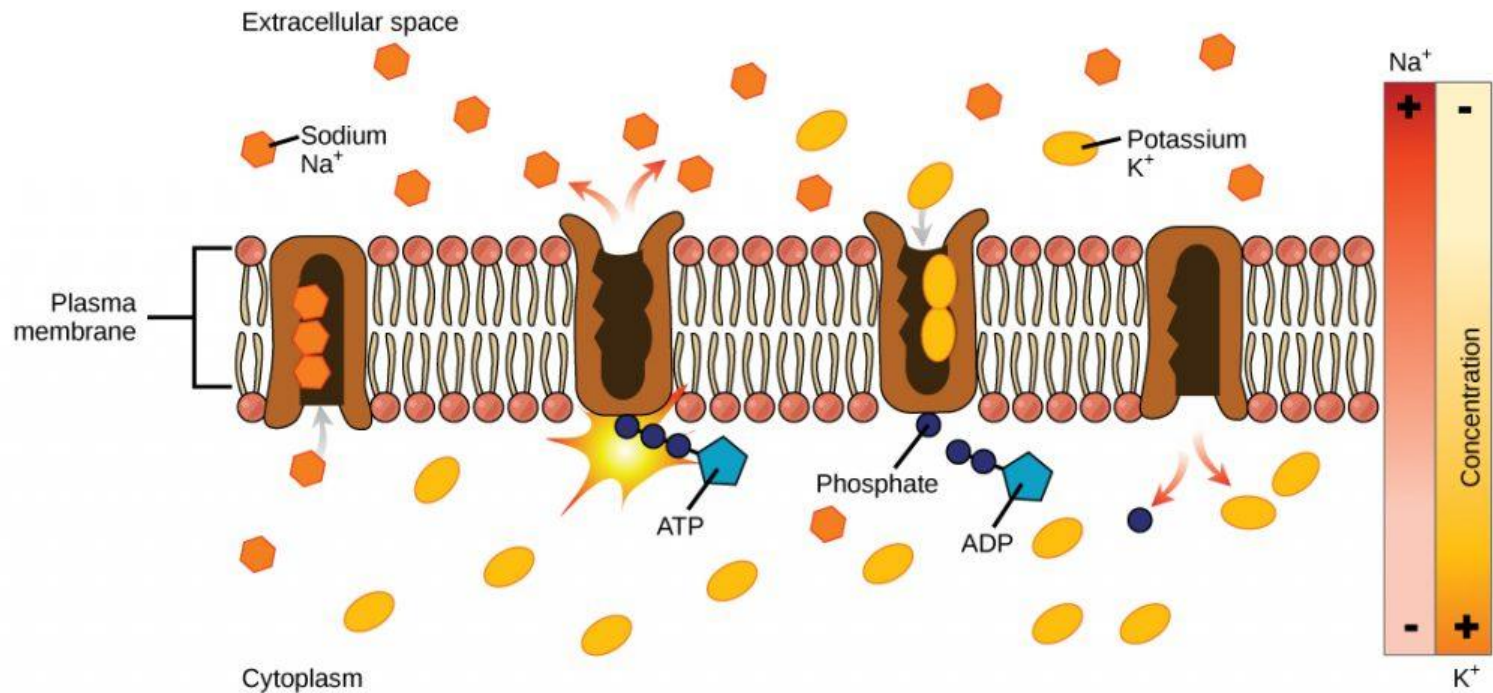
minimum energy required to cause a reaction to occur



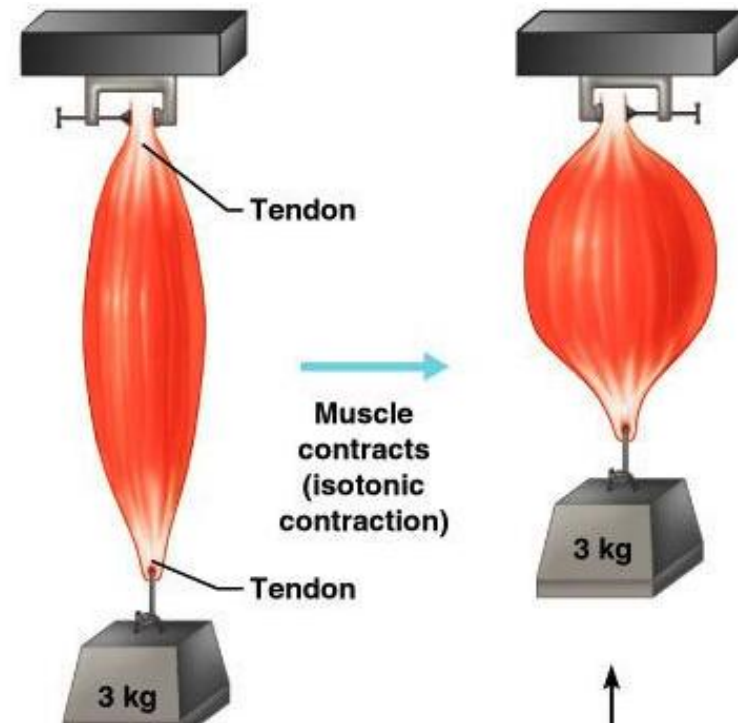
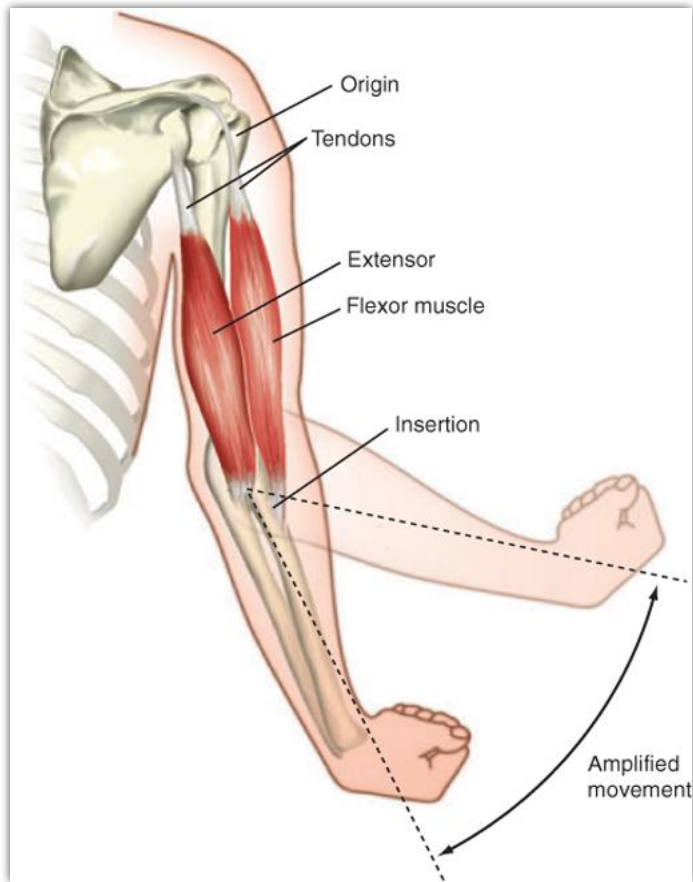
Transporters

ATP hydrolysis (release of energy)

Transform structure of channels



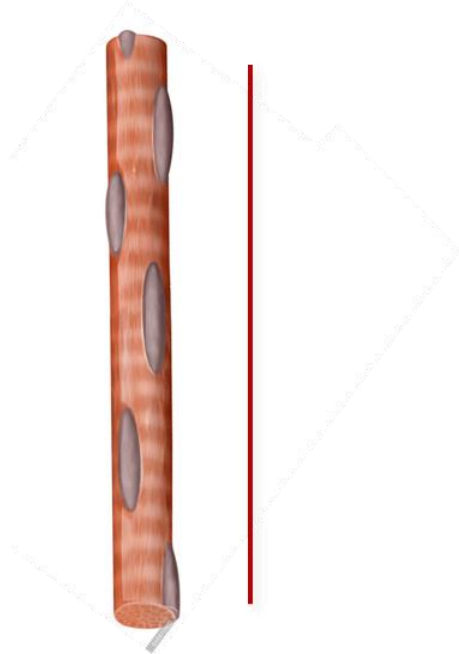
Movement (physical reaction)



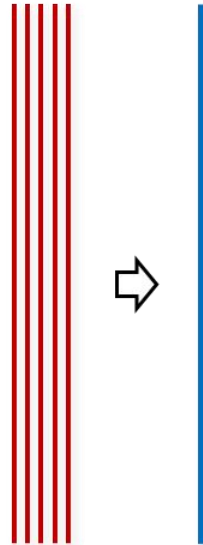
Organization of Skeletal Muscle



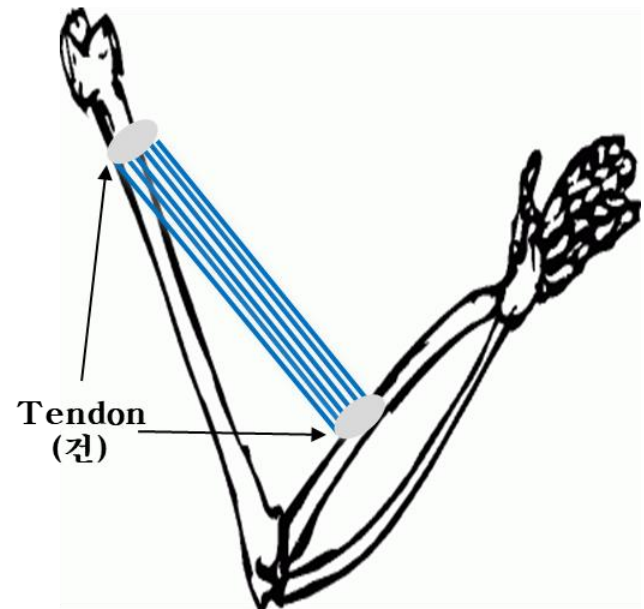
Muscle Fiber
(근섬유, 근세포)



Fasciculus
(근다발)



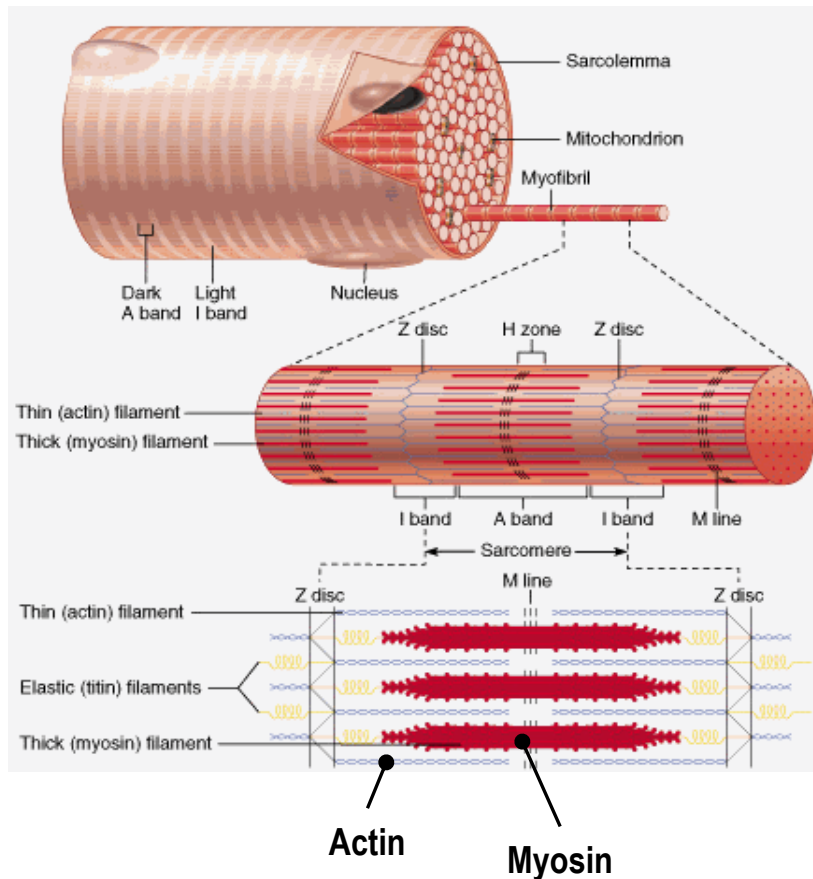
Skeletal muscle
(골격근)



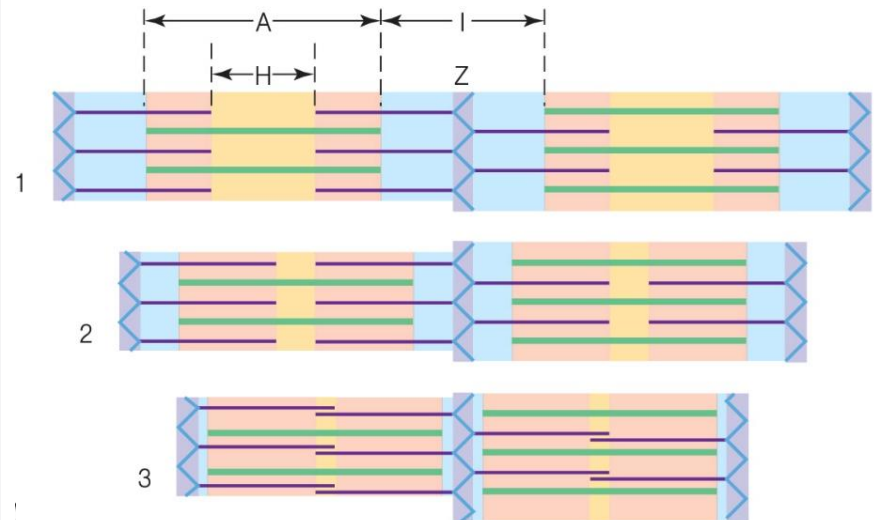
Organization of Skeletal Muscle



Muscle cell



Muscle contraction



Organization of Skeletal Muscle

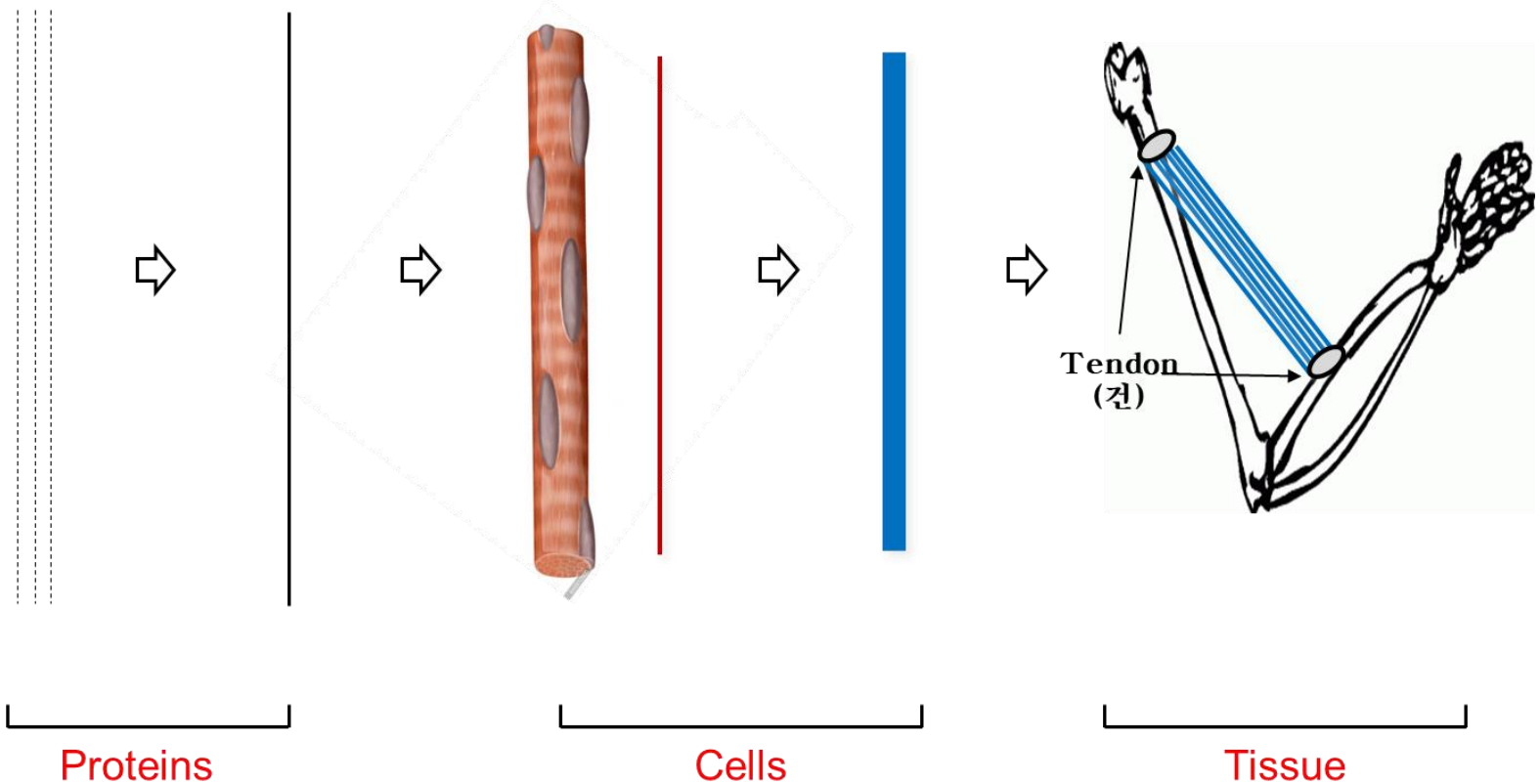
Myofilament
(근필라멘트)

Myofibril
(근원섬유)

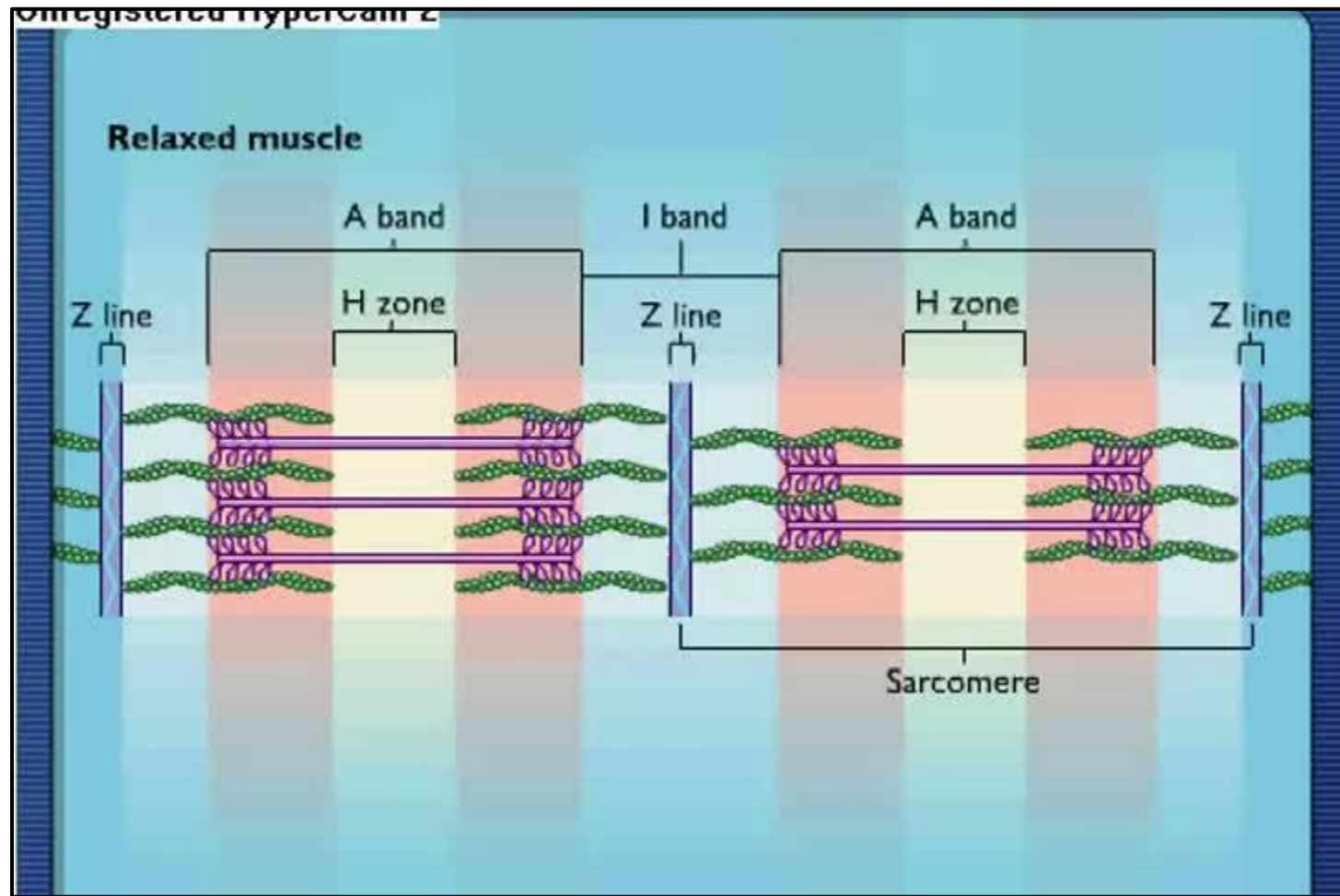
Muscle Fiber
(근섬유, 근세포)

Fascicle
(근다발)

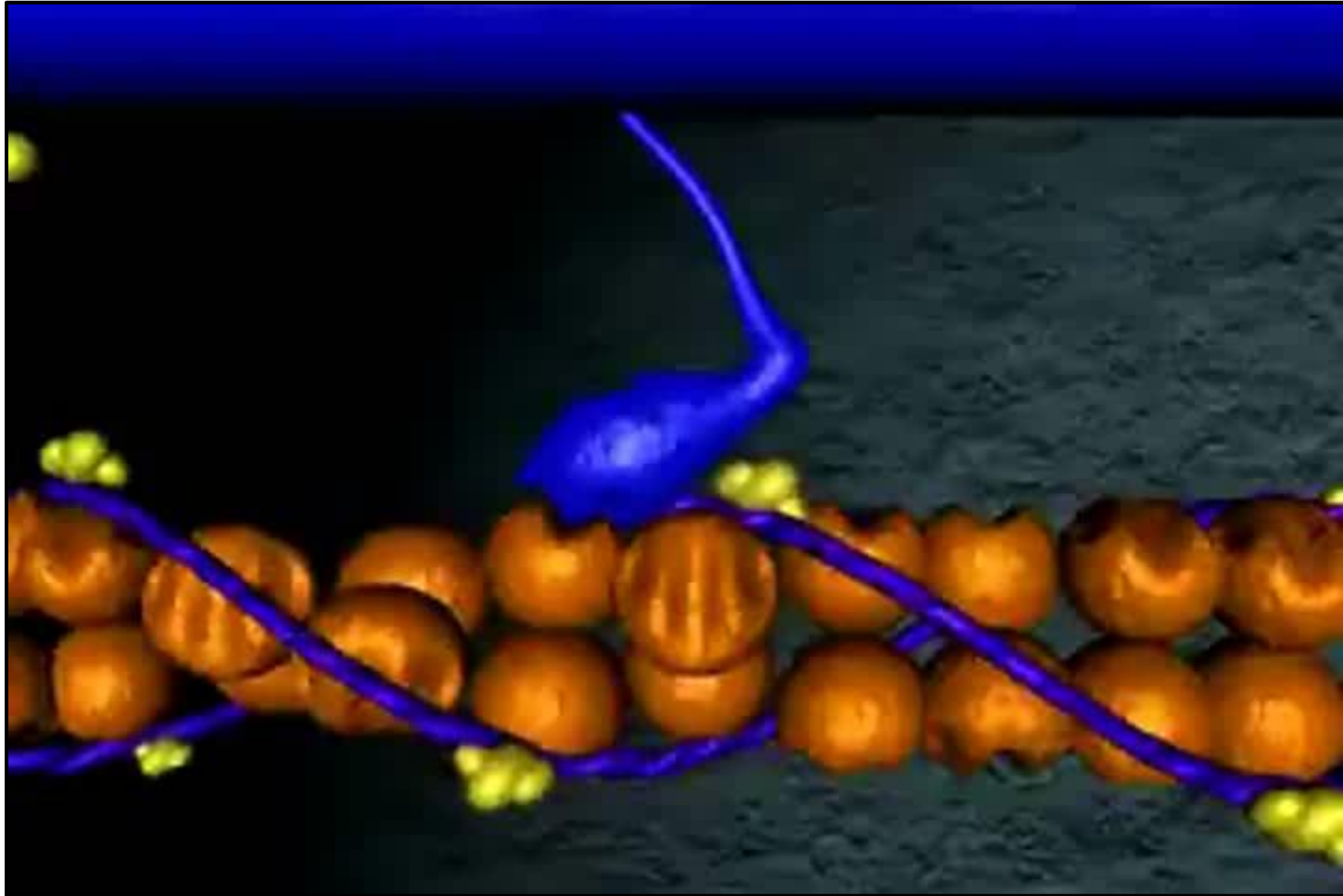
Skeletal muscle
(골격근)



Movement (physical reaction)



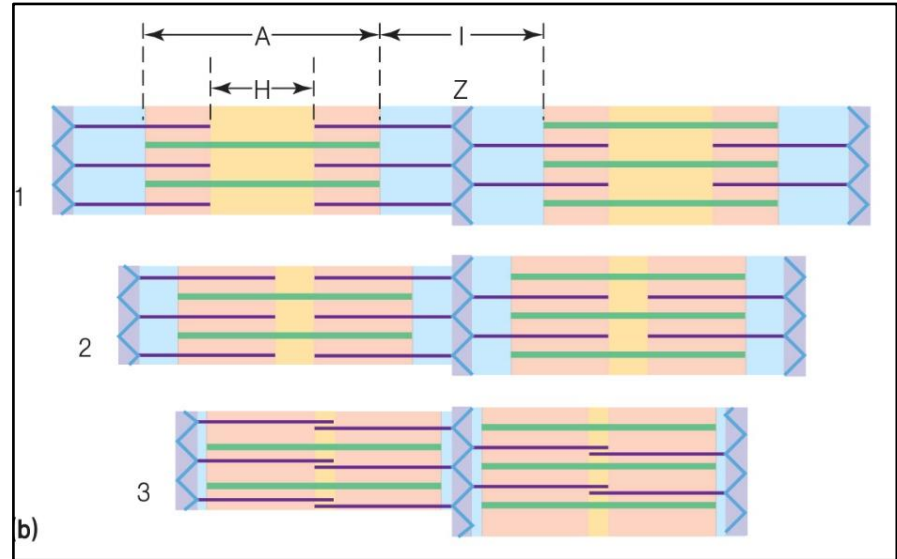
Movement (physical reaction)



Movement (physical reaction)

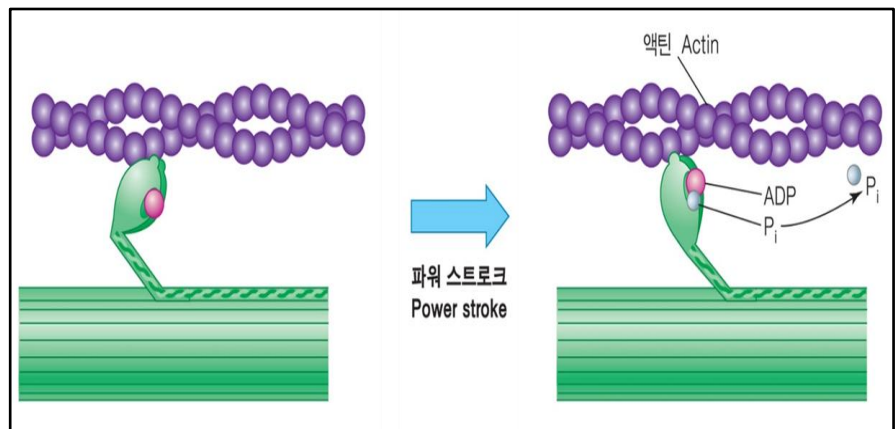
Muscle contraction

Shortening of sarcomere =>
shortening of muscle length

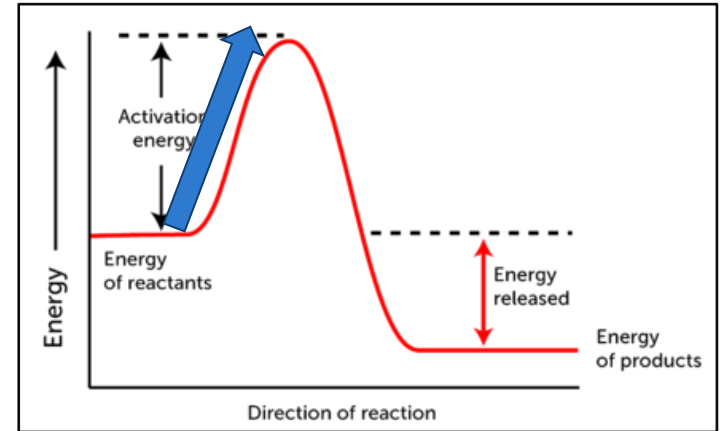
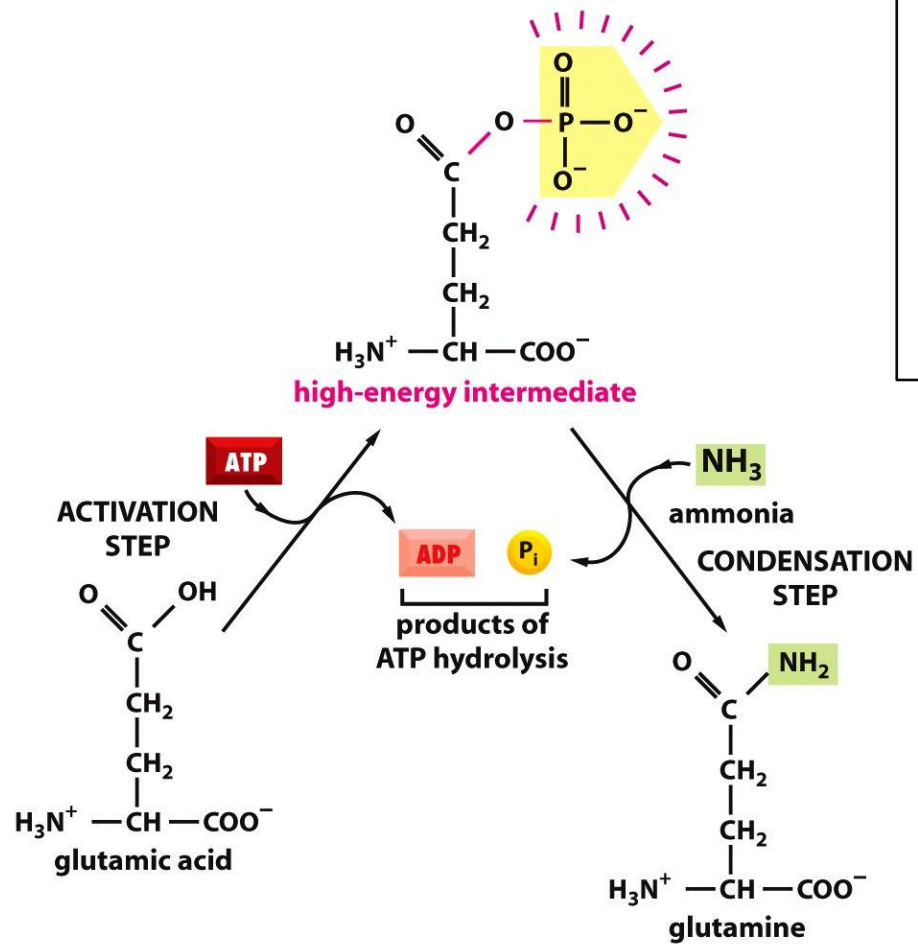


Power stroke

ATP hydrolysis changes structure
of cross-bridge (protein)



Substrate activation

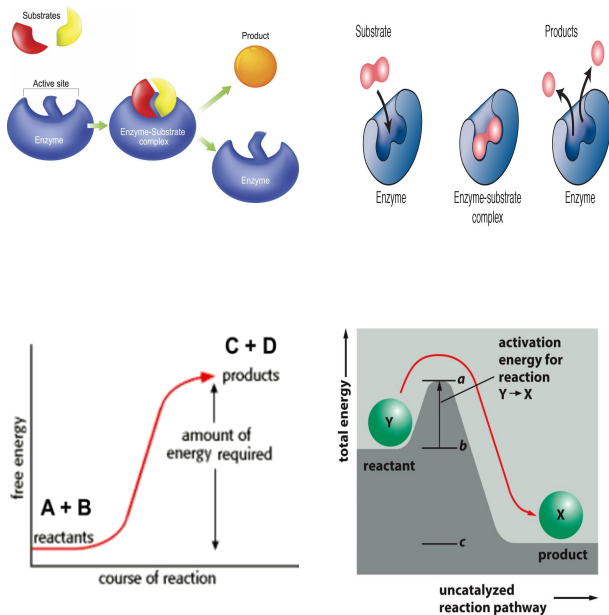


Use of ATP as energy

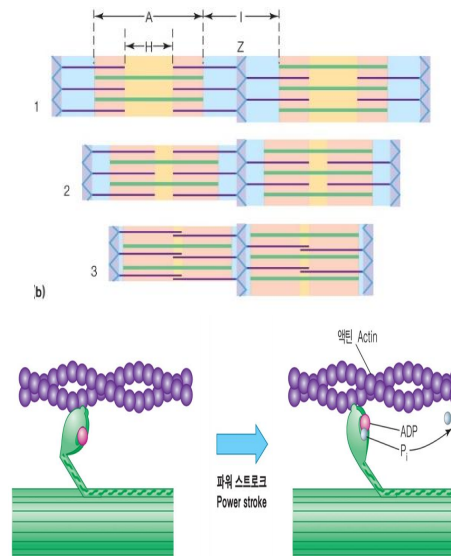
ATP hydrolysis (release of energy)

- Transform protein (enzyme, transporter, myosin, etc.) structure
- Remove activation energy

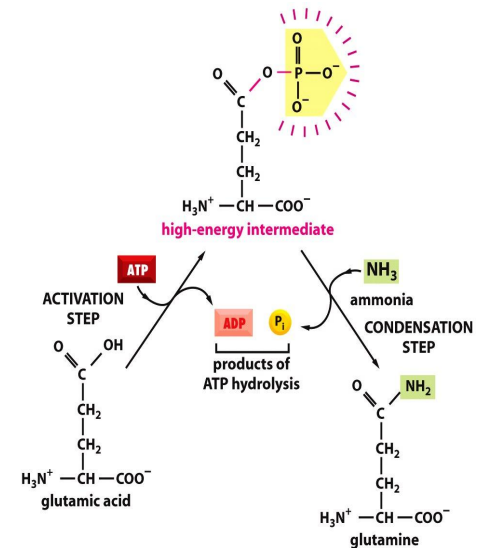
Enzyme



Protein



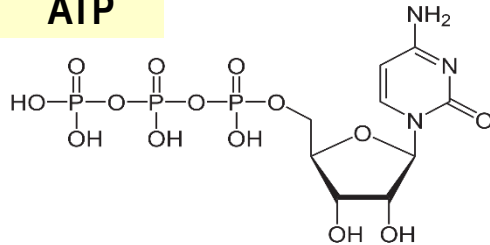
Substrate activation



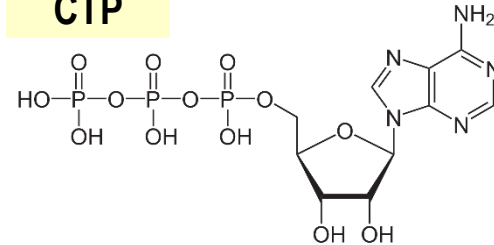
Universal biological energy ATP



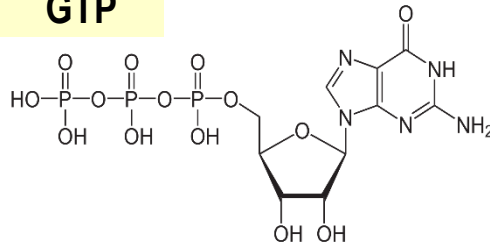
ATP



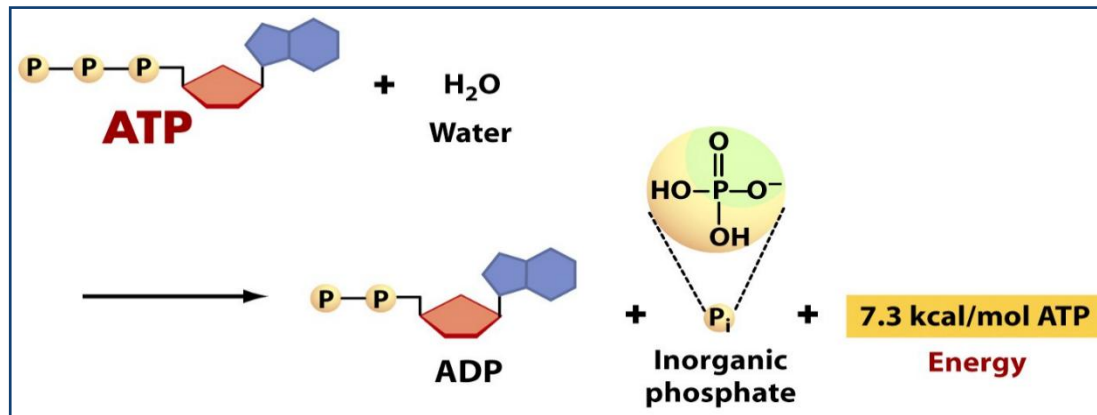
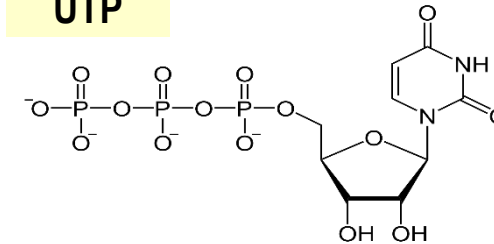
CTP

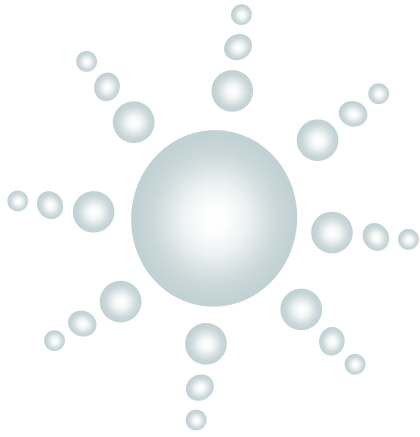


GTP



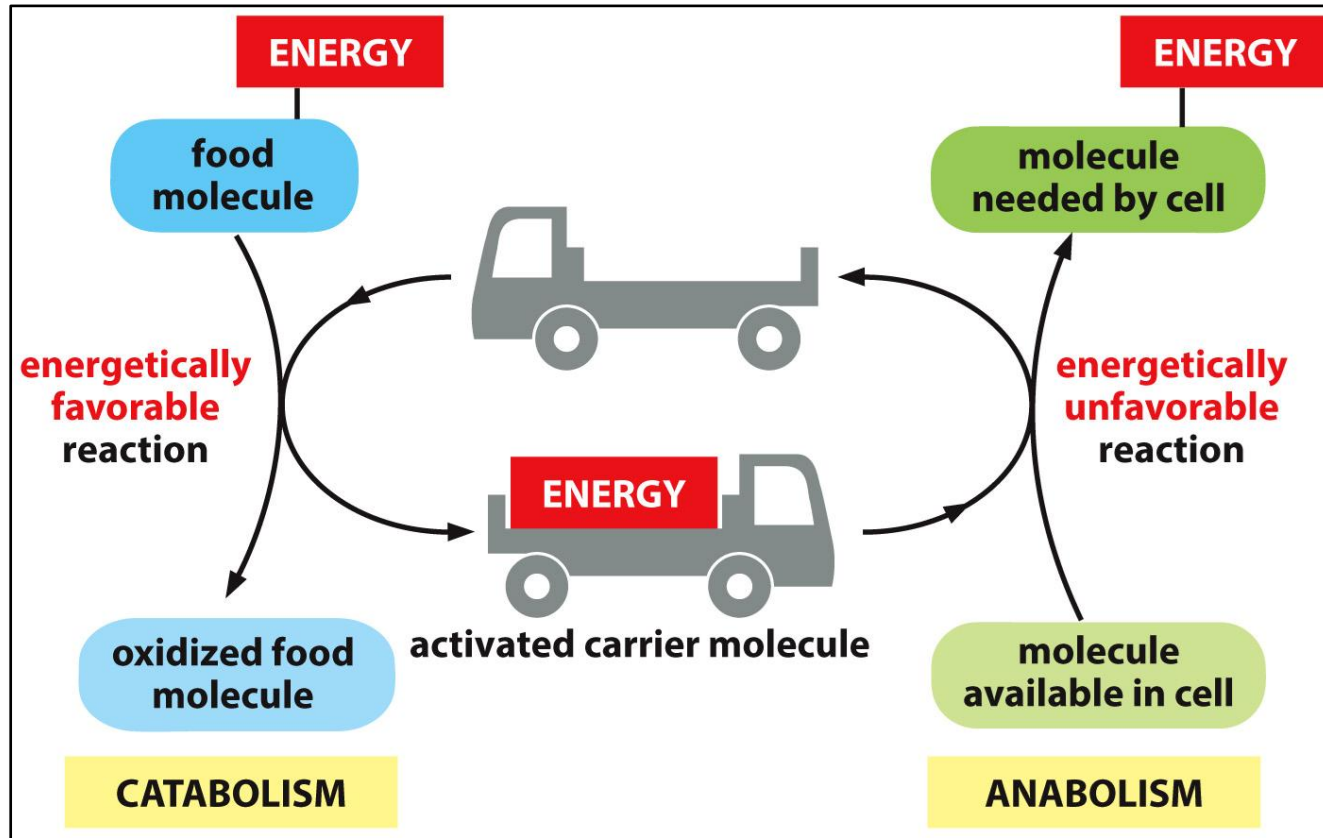
UTP





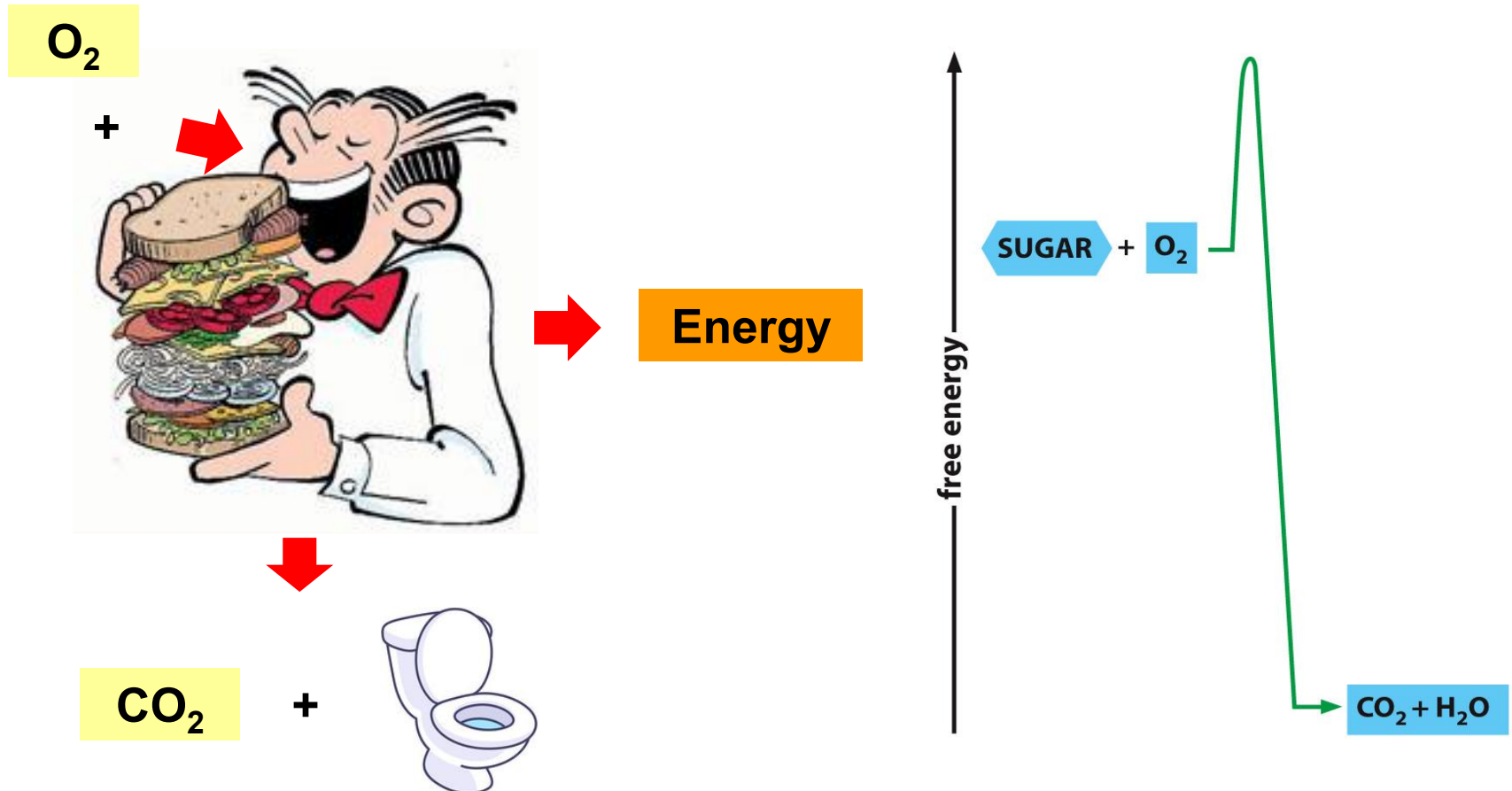
How ATP is generated?

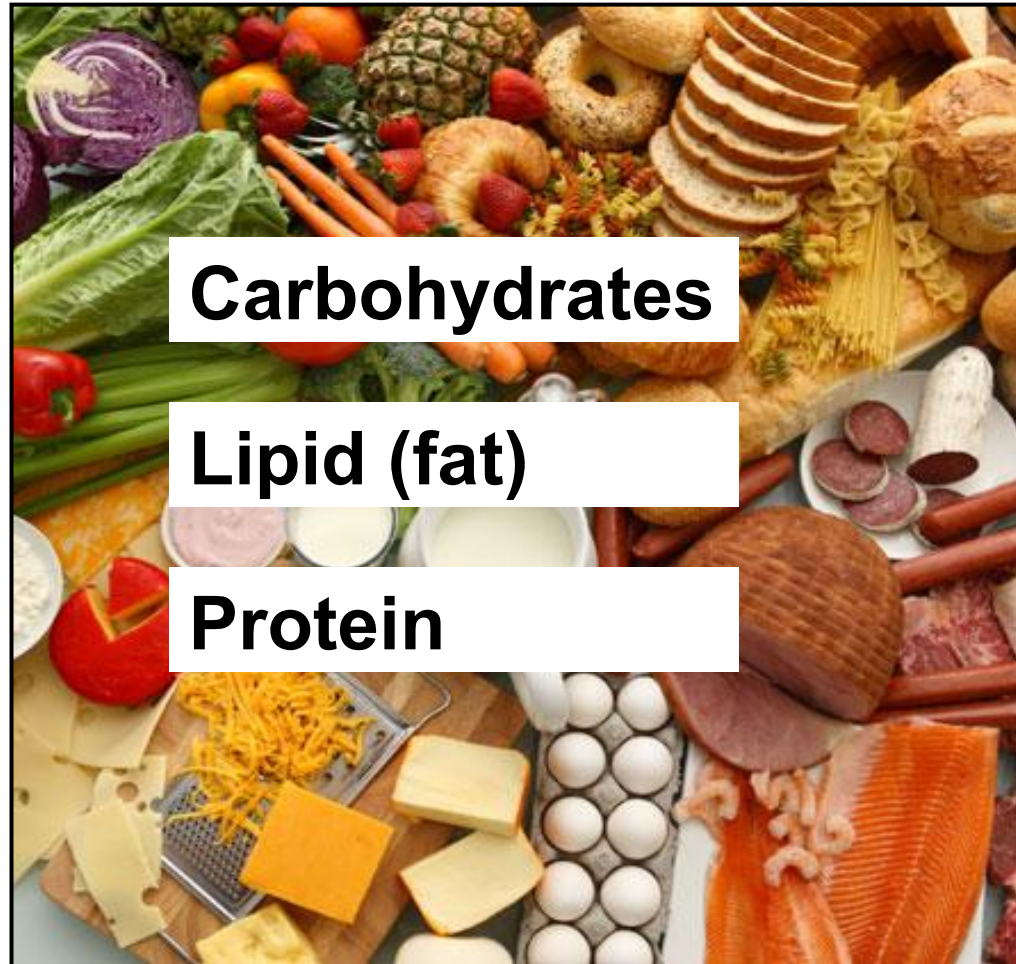
Energy conversion



Energy conversion

Extract energy from complex molecules (Food) : Catabolism
Produce cellular energy currency (ATP) : Energy metabolism



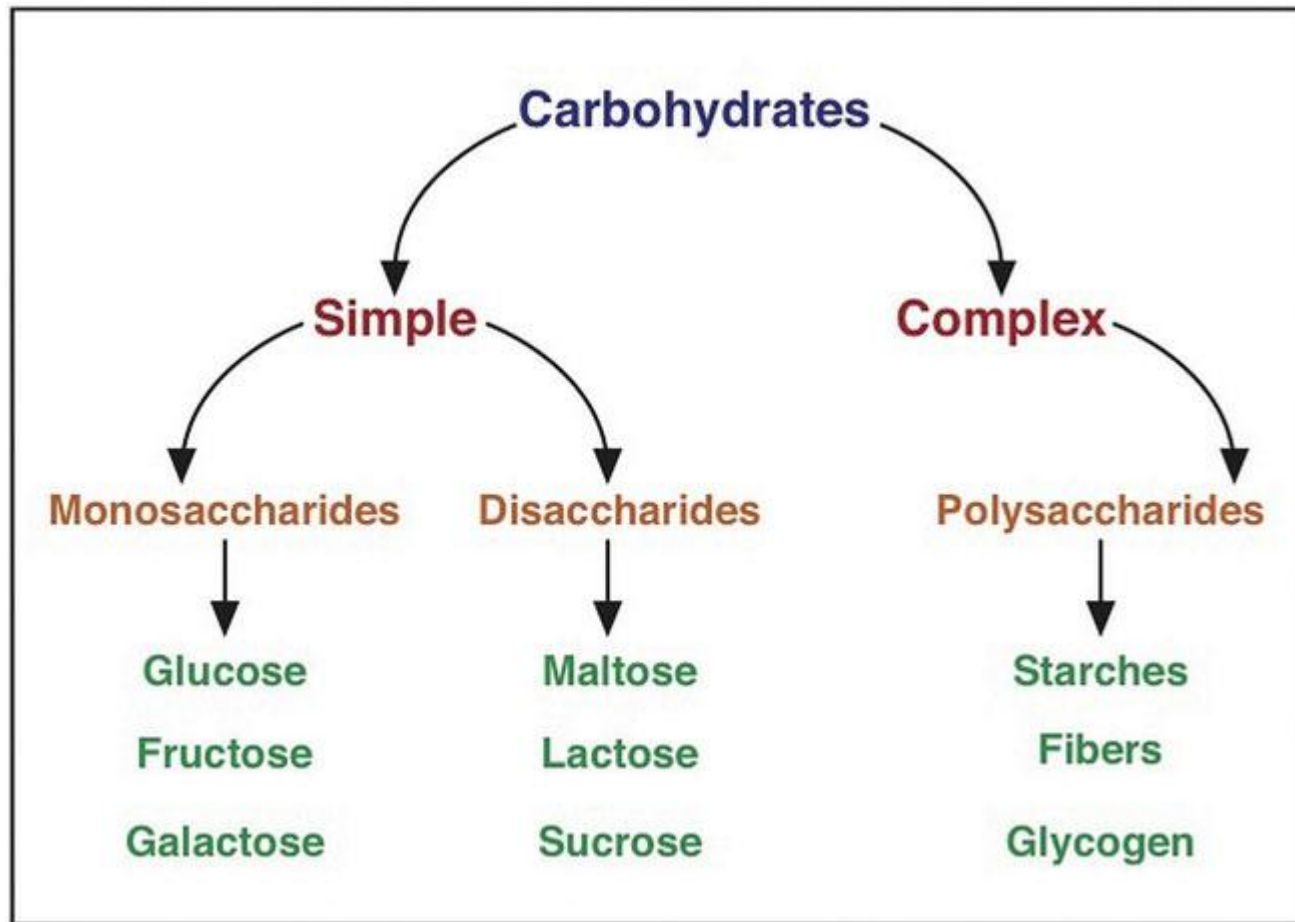


Carbohydrates

Lipid (fat)

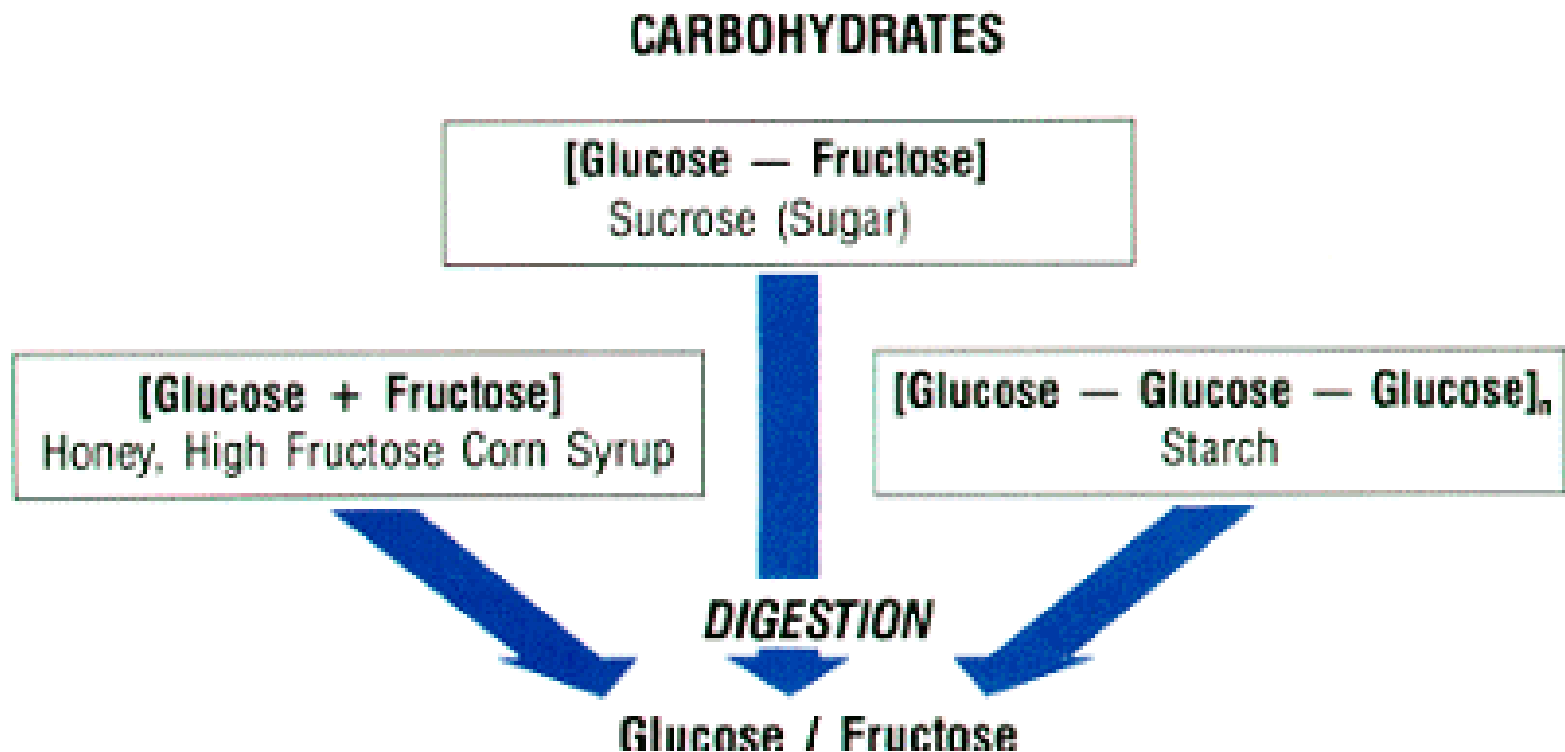
Protein

Carbohydrates



Carbohydrates in food

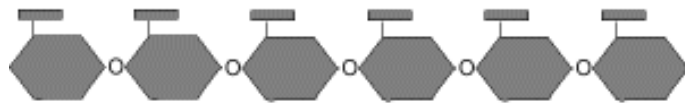
- Carbohydrates are absorbed as glucose, galactose, and fructose
- Hepatic conversion : Galactose $\rightarrow$ glucose,
Fructose $\rightarrow$ glucose, glycogen, or lipids



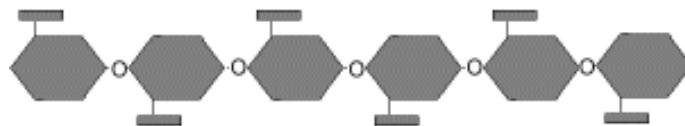
Carbohydrates



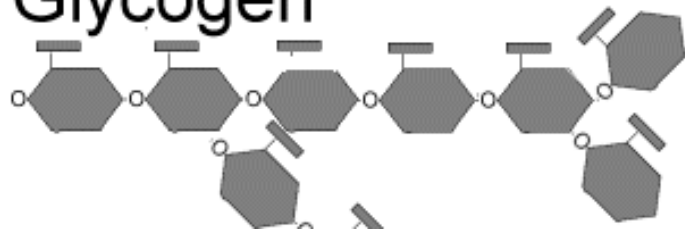
Starch



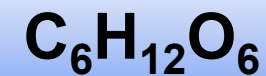
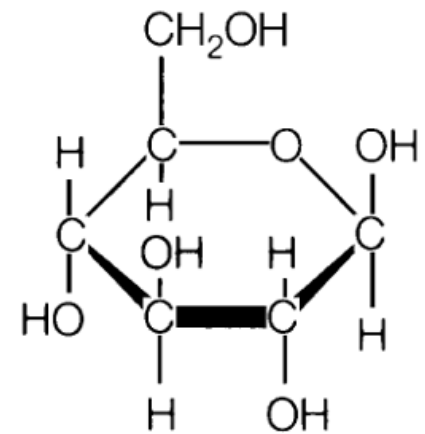
Cellulose



Glycogen

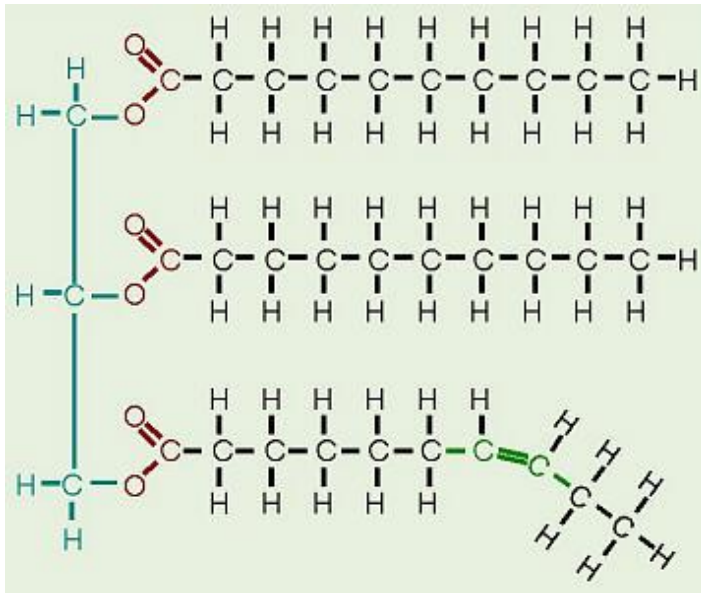


Glucose

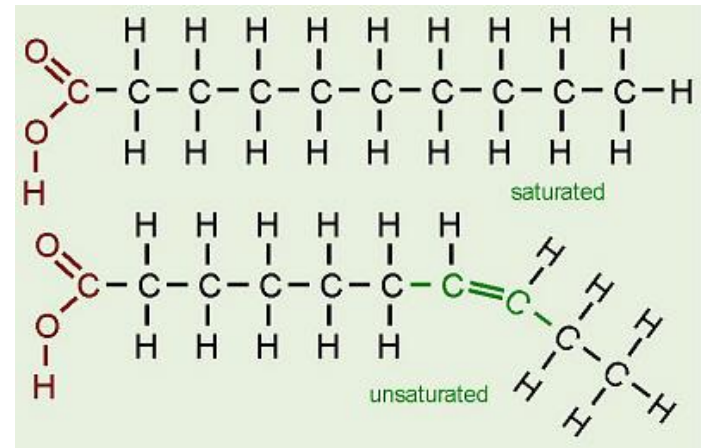


Lipid (Fat)

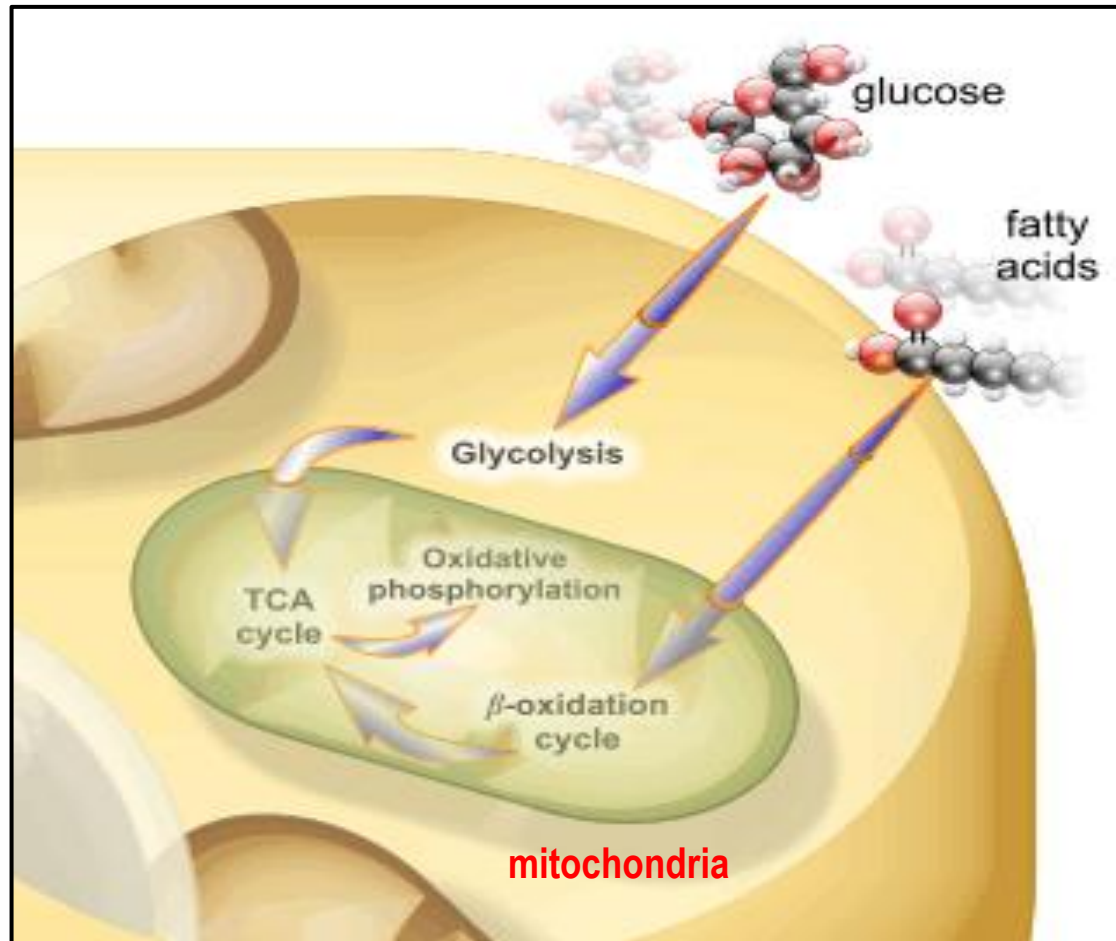
- Fat : glycerol backbone and fatty acids
- Fatty acids : carboxyl group at the end, variable length of hydrocarbon chains (saturated and unsaturated)

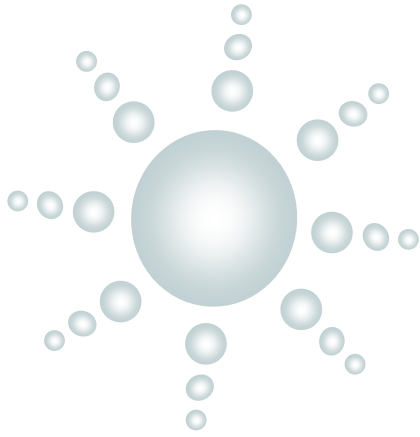


Fatty acids



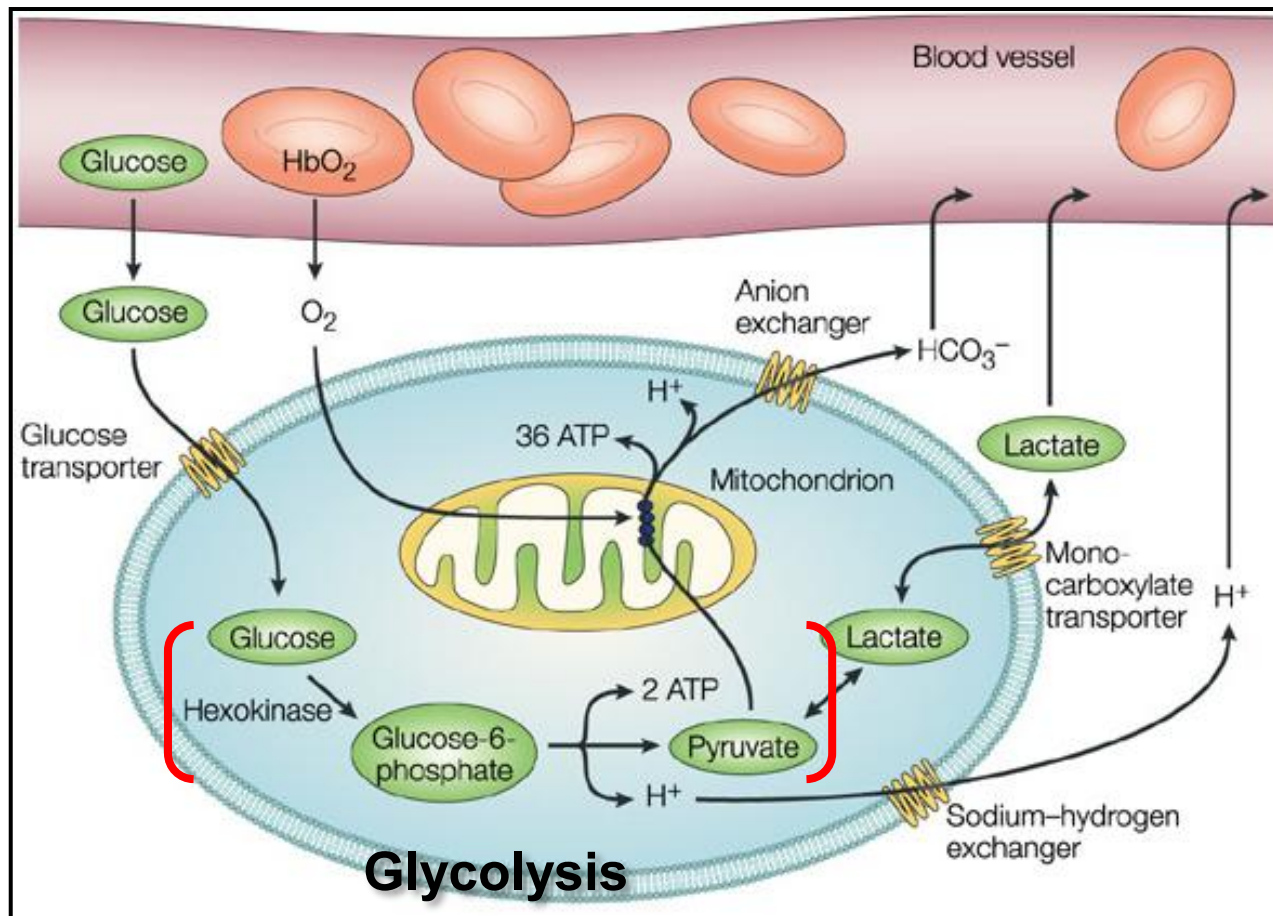
Energy metabolism





ATP Production from Glucose

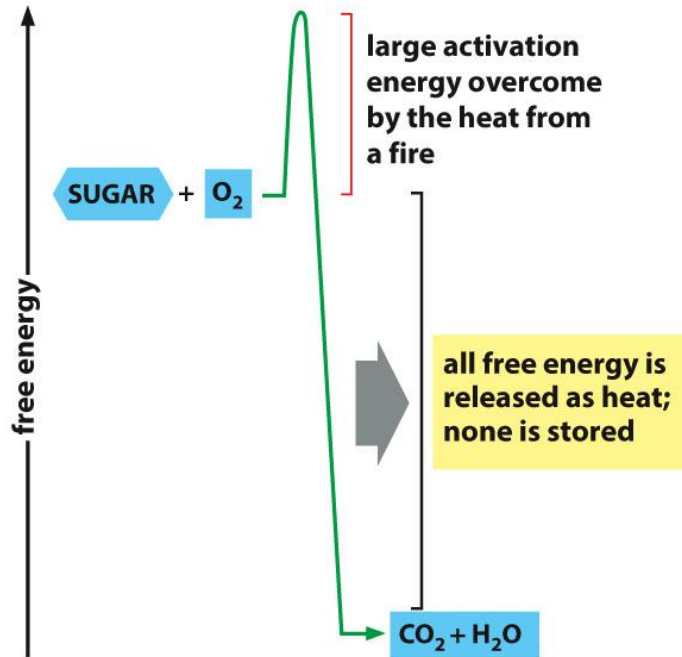
Glucose metabolism for ATP generation



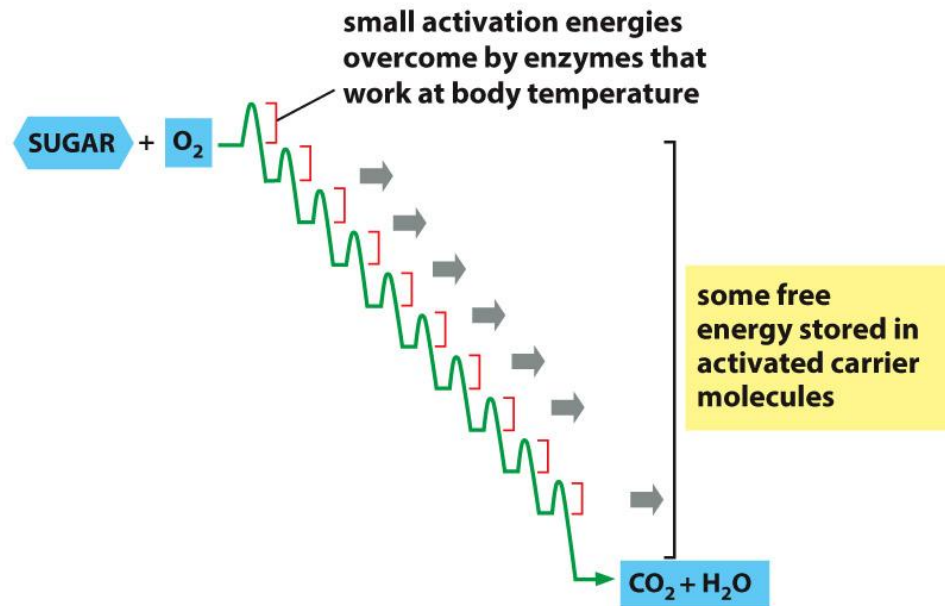
Energy conversion



(A) DIRECT BURNING OF SUGAR
IN NONLIVING SYSTEM

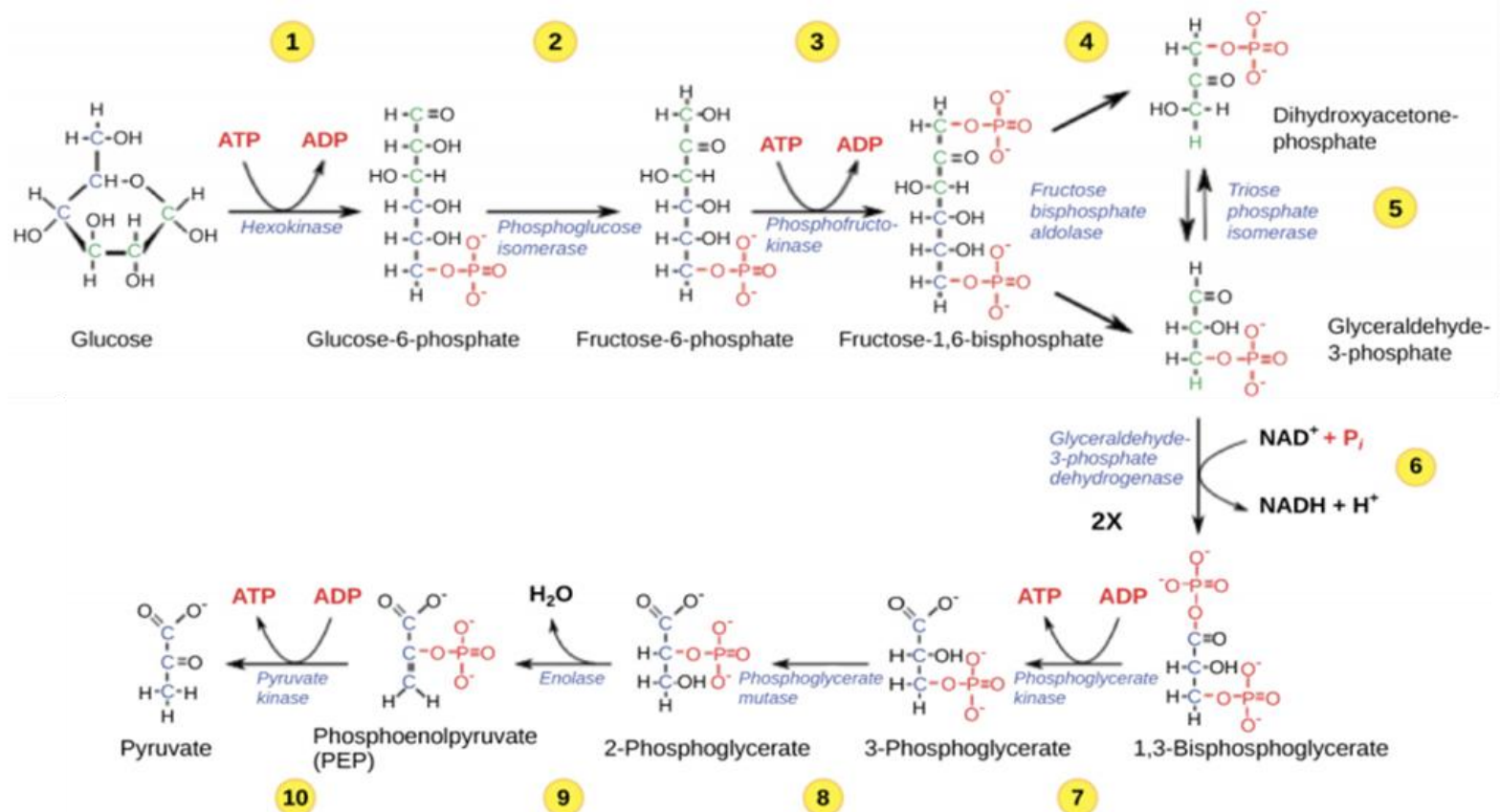


(B) STEPWISE OXIDATION OF SUGAR IN CELLS

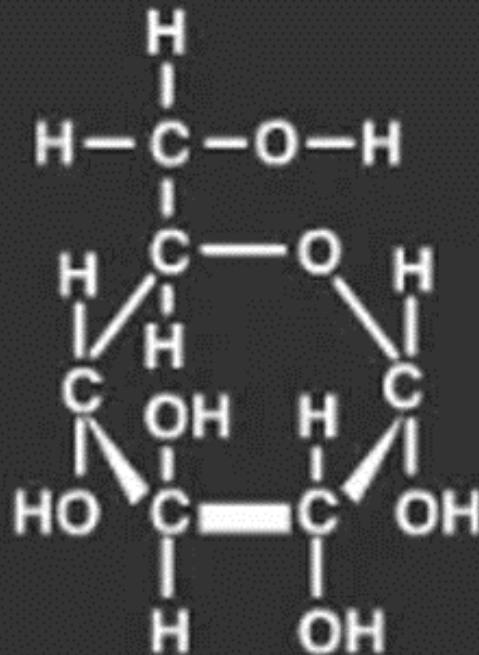


Glycolysis

An oxygen-independent, ten-step cytosolic pathway that converts a single glucose molecule into two pyruvate molecules, producing ATP and NADH

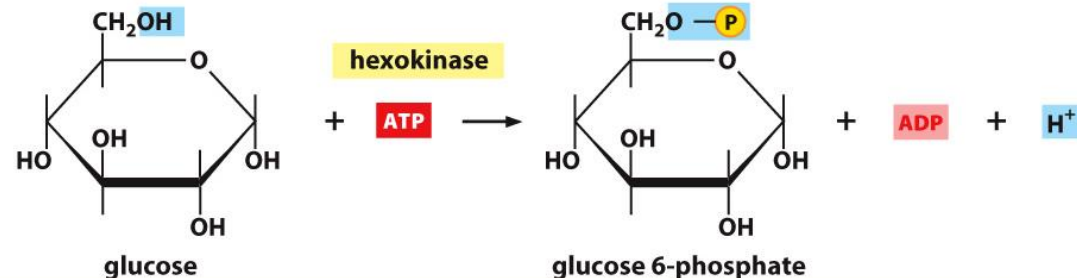


Glycolysis

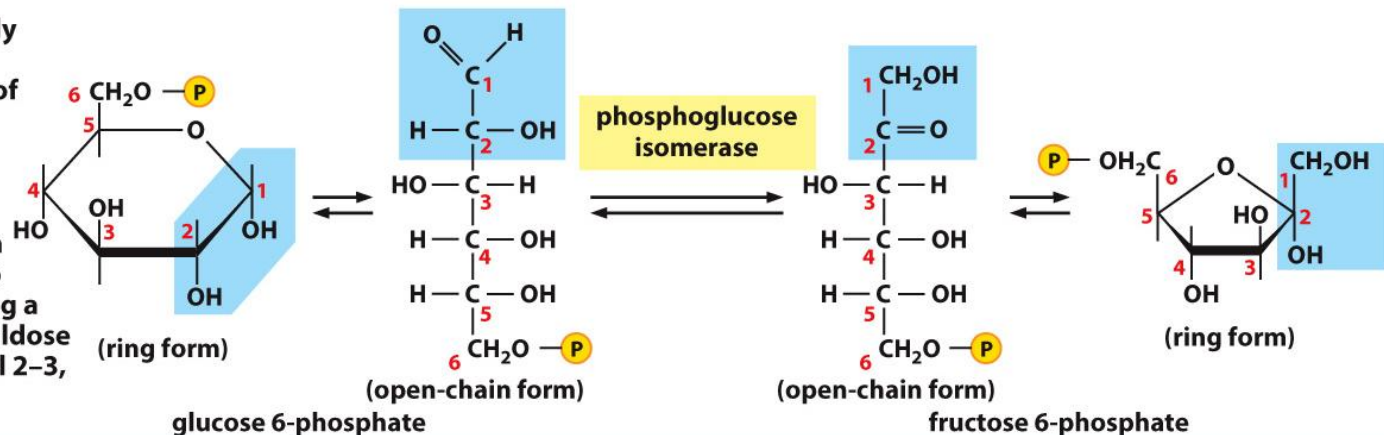


Glycolysis

Step 1 Glucose is phosphorylated by ATP to form a sugar phosphate. The negative charge of the phosphate prevents passage of the sugar phosphate through the plasma membrane, trapping glucose inside the cell.



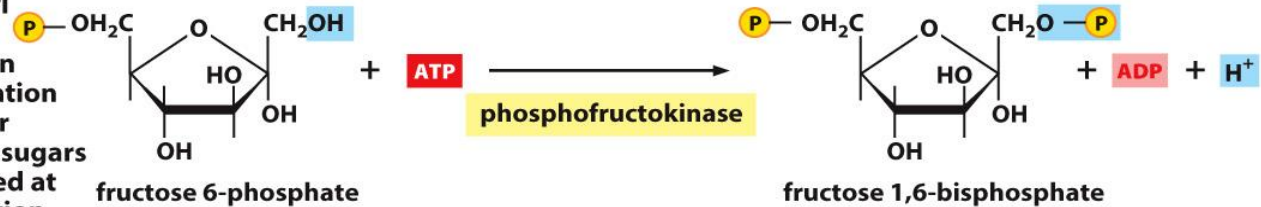
Step 2 A readily reversible rearrangement of the chemical structure (isomerization) moves the carbonyl oxygen from carbon 1 to carbon 2, forming a ketose from an aldose sugar. (See Panel 2-3, pp. 70-71.)



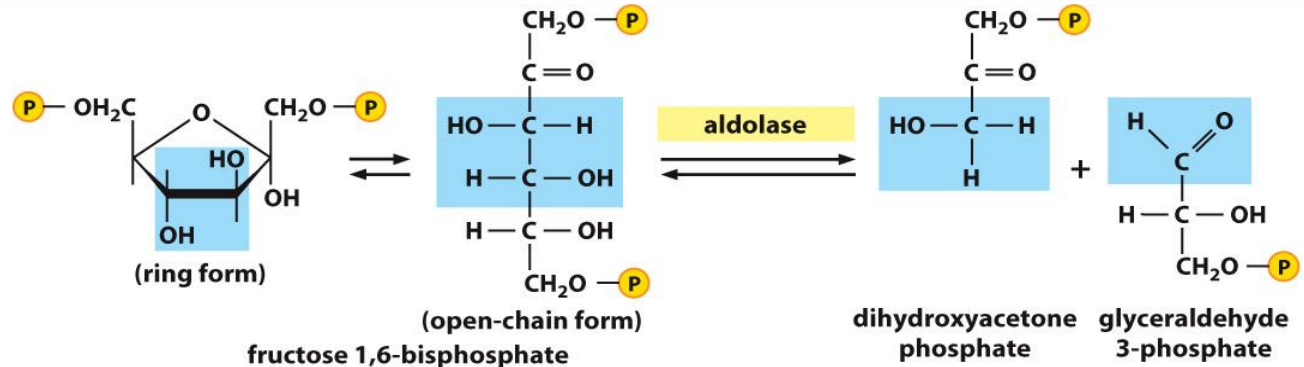
Glycolysis



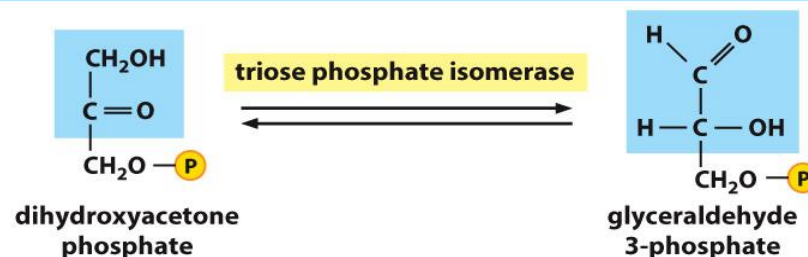
Step 3 The new hydroxyl group on carbon 1 is phosphorylated by ATP, in preparation for the formation of two three-carbon sugar phosphates. The entry of sugars into glycolysis is controlled at this step, through regulation of the enzyme *phosphofructokinase*.



Step 4 The six-carbon sugar is cleaved to produce two three-carbon molecules. Only the glyceraldehyde 3-phosphate can proceed immediately through glycolysis.



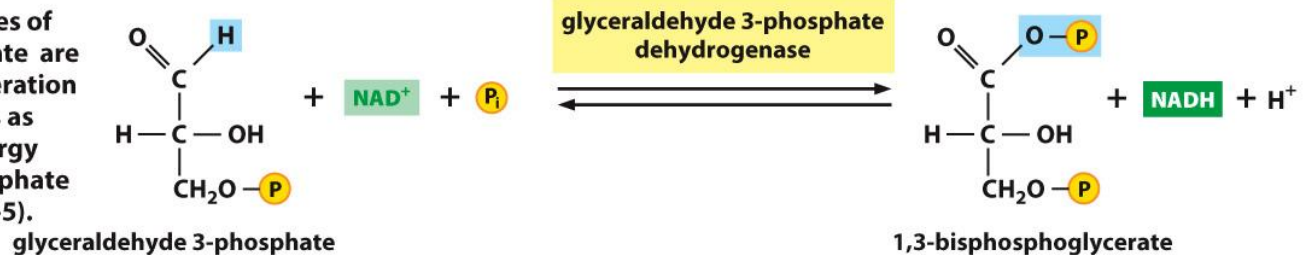
Step 5 The other product of step 4, dihydroxyacetone phosphate, is isomerized to form glyceraldehyde 3-phosphate.



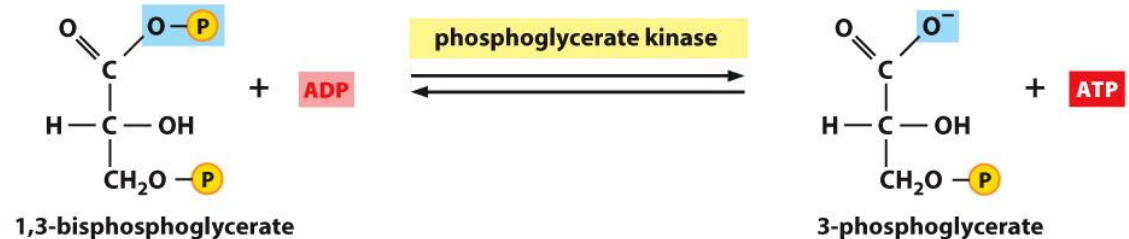
Glycolysis



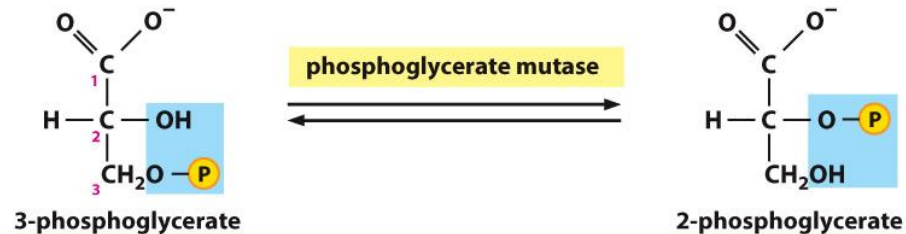
Step 6 The two molecules of glyceraldehyde 3-phosphate are oxidized. The energy-generation phase of glycolysis begins, as NADH and a new high-energy anhydride linkage to phosphate are formed (see Figure 13-5).



Step 7 The transfer to ADP of the high-energy phosphate group that was generated in step 6 forms ATP.



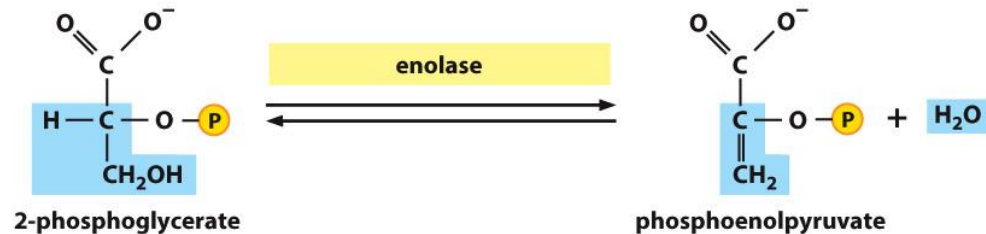
Step 8 The remaining phosphate ester linkage in 3-phosphoglycerate, which has a relatively low free energy of hydrolysis, is moved from carbon 3 to carbon 2 to form 2-phosphoglycerate.



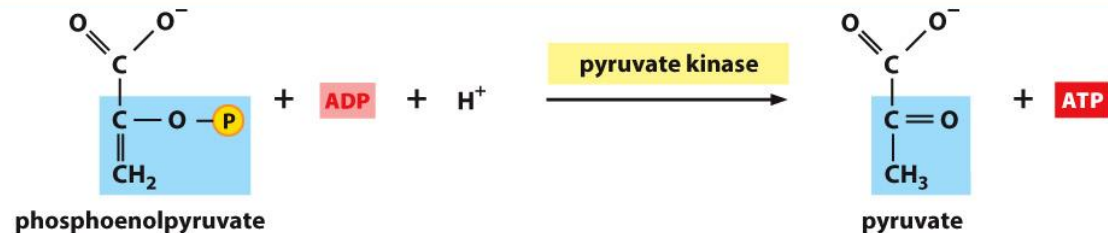
Glycolysis



Step 9 The removal of water from 2-phosphoglycerate creates a high-energy enol phosphate linkage.



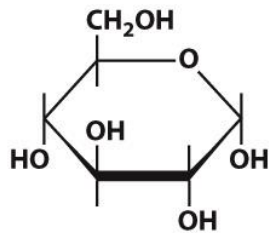
Step 10 The transfer to ADP of the high-energy phosphate group that was generated in step 9 forms ATP, completing glycolysis.



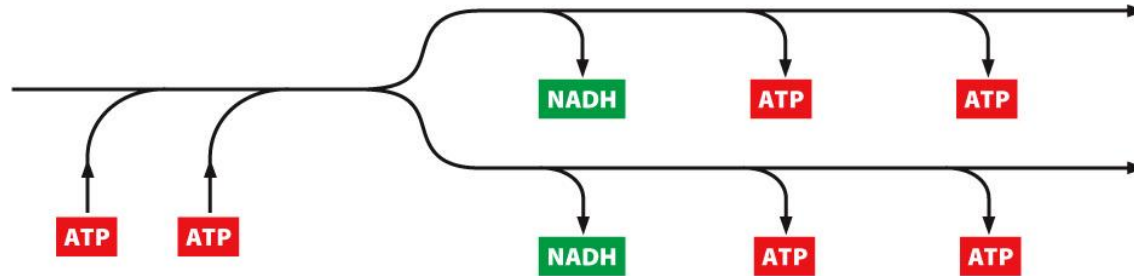
Glycolysis



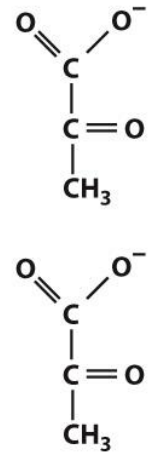
NET RESULT OF GLYCOLYSIS



glucose



In addition to the pyruvate, the net products are two molecules of ATP and two molecules of NADH.



two molecules of pyruvate



Glycolysis

